Seismological Society of America

SSA 2026 Workshop

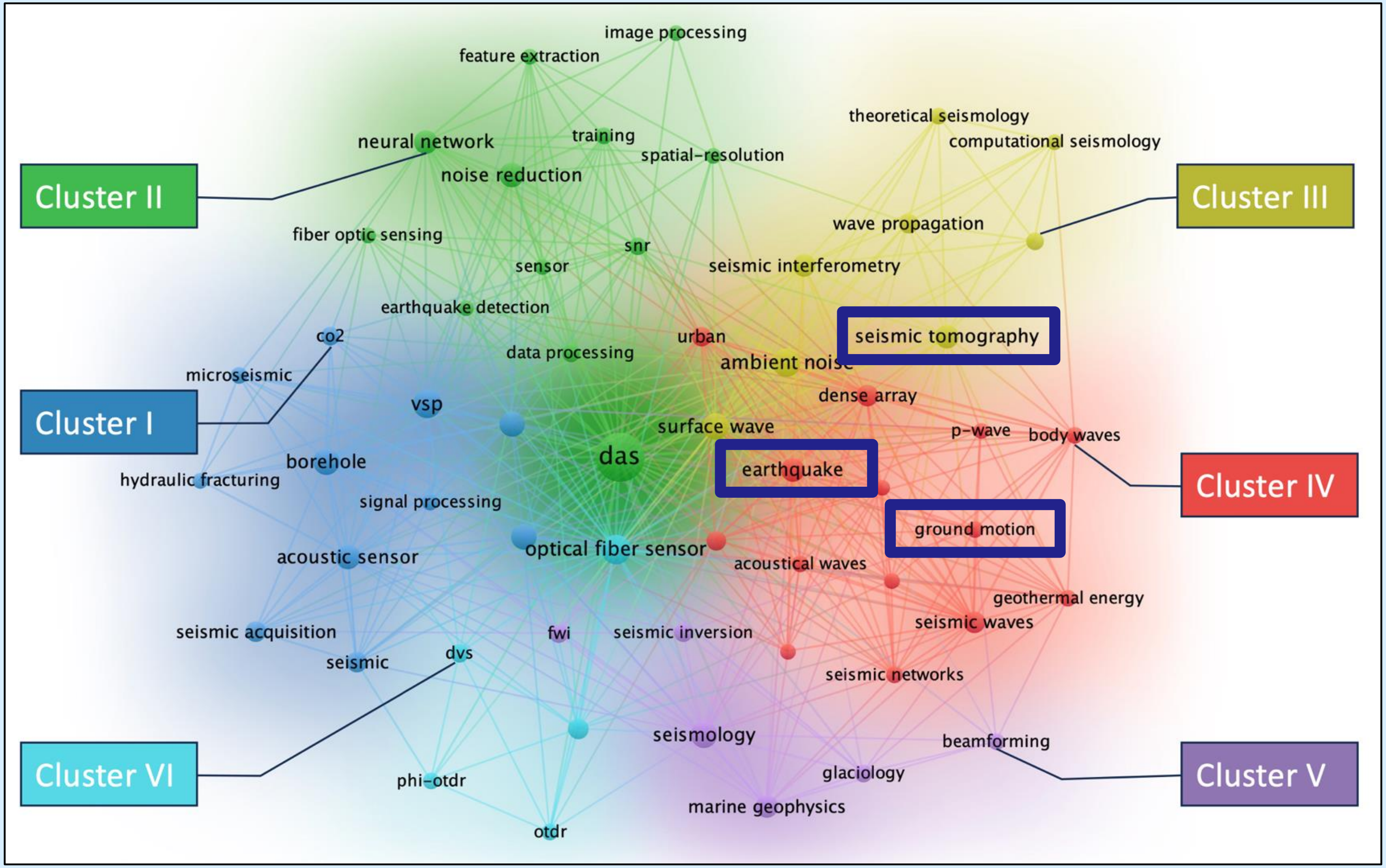
Introduction to DAS for Seismology: From Data Acquisition to Analysis

Eikonal Traveltime Tomography with DAS

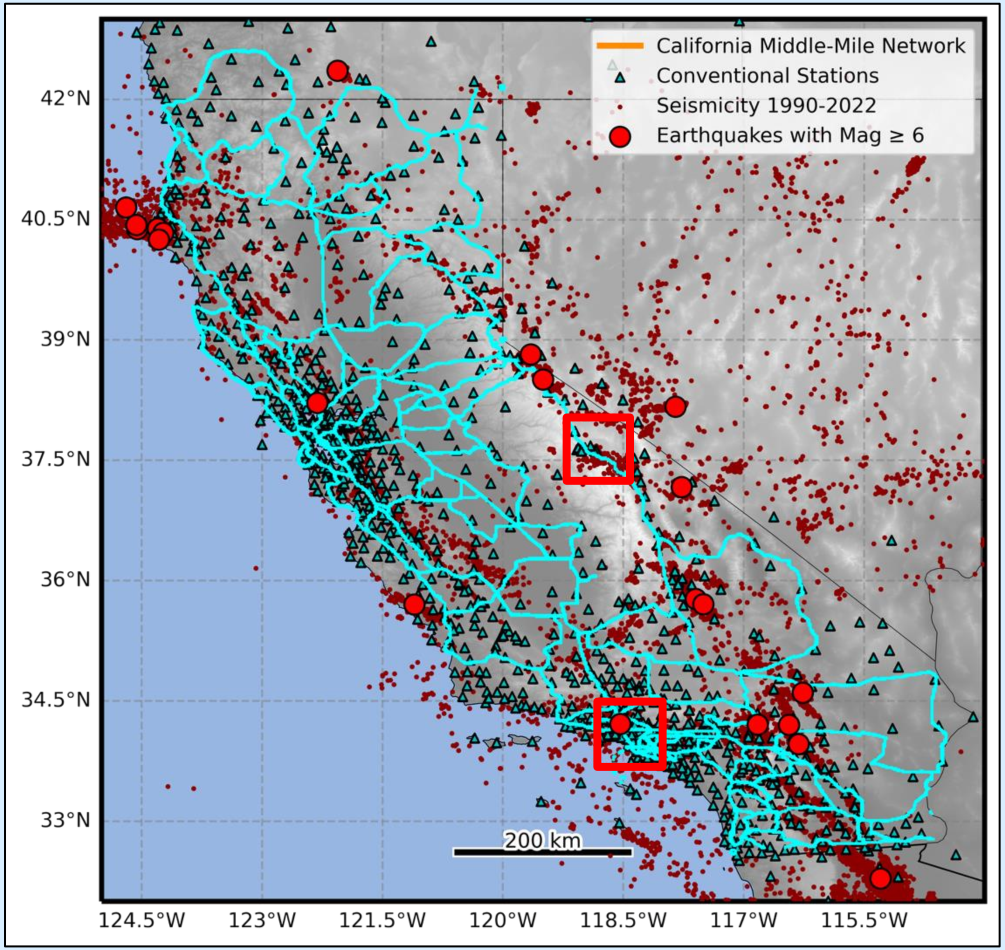
Ettore Biondi*, Jiaxuan Li, Chun Zhang, and Weiqiang Zhu
(alphabetical order)

April 14th, 2026

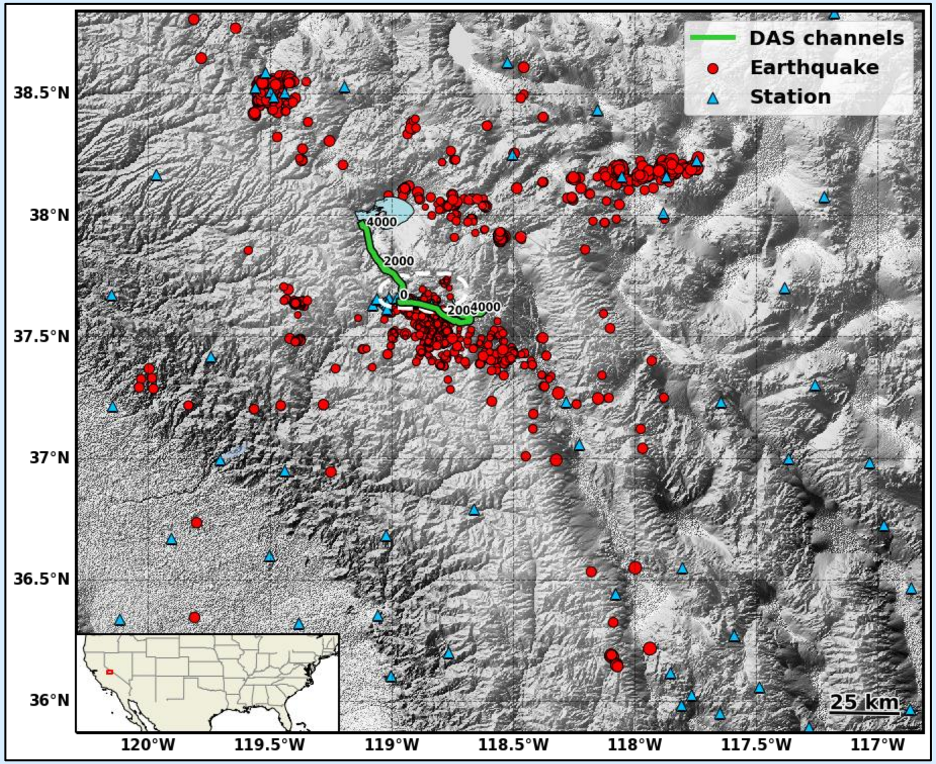
Distributed acoustic sensing (DAS)



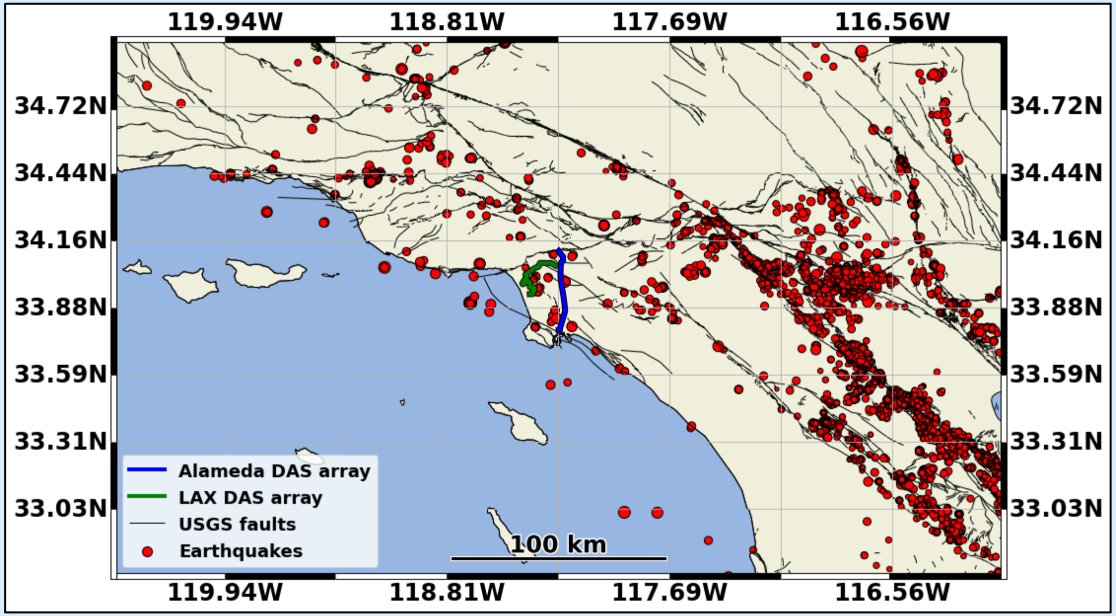
Distributed acoustic sensing (DAS)



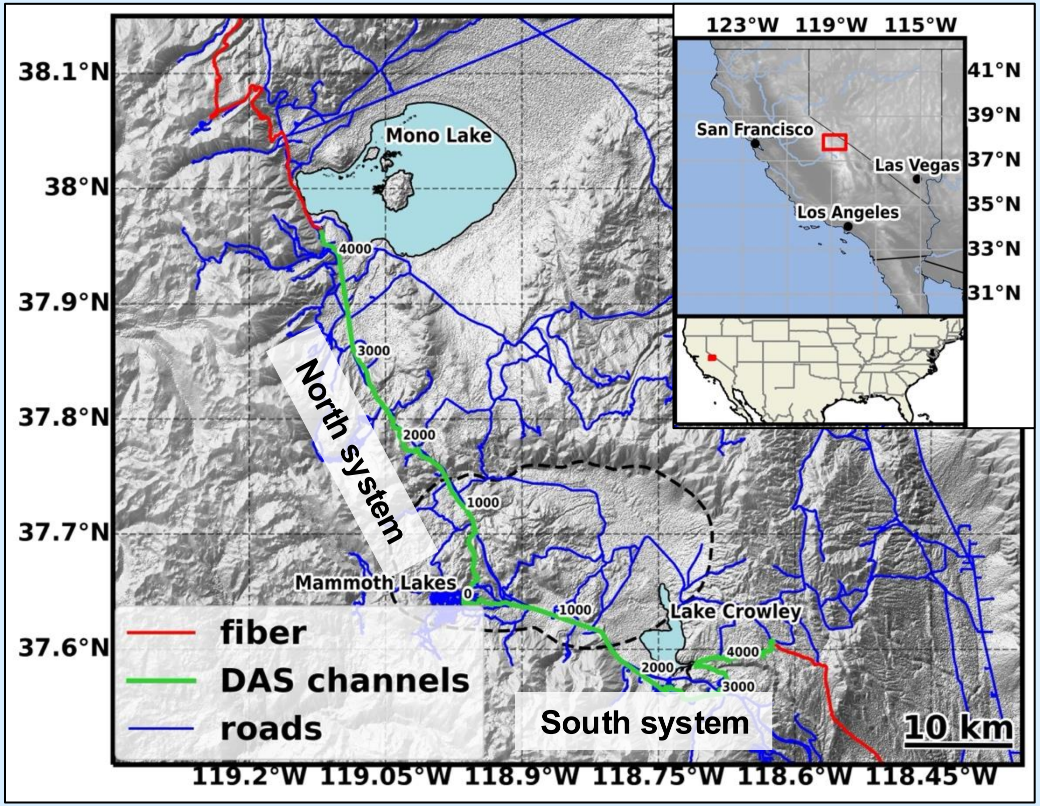
The Long Valley volcanic field

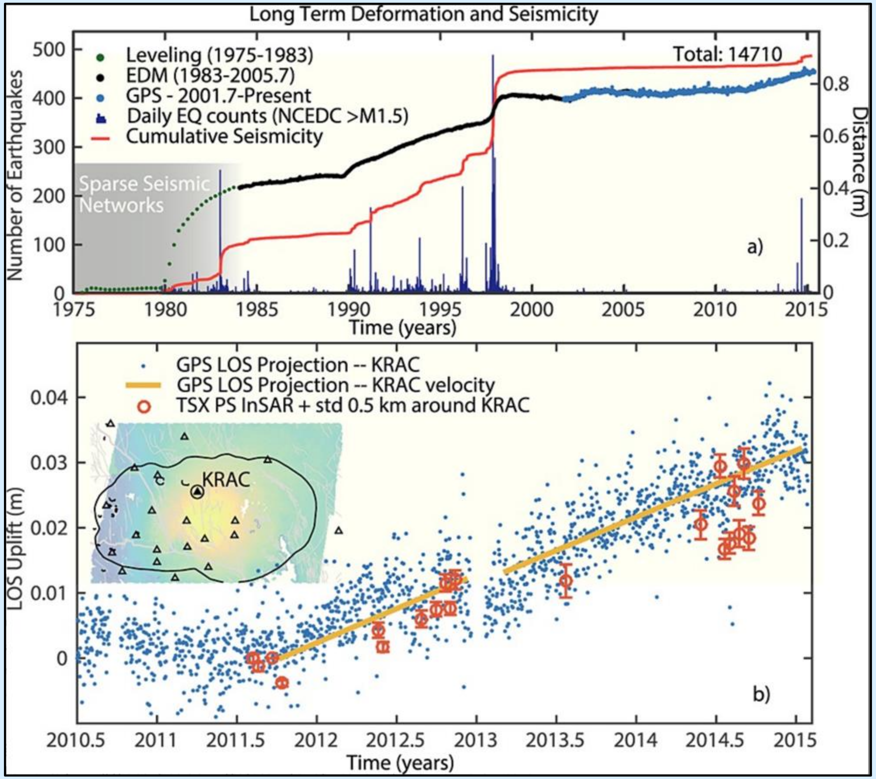


Los Angeles sedimentary basin



The Long Valley supervolcano

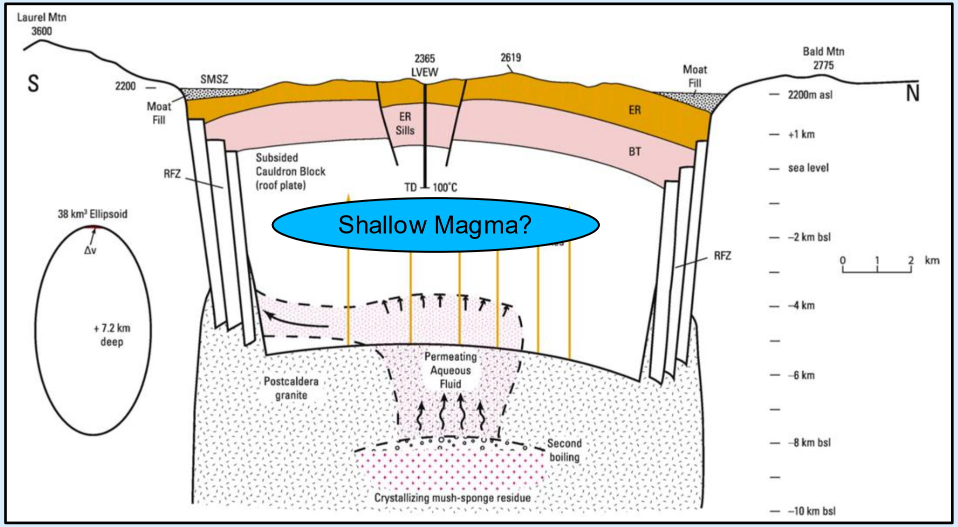




Montgomery-Brown et al. 2015

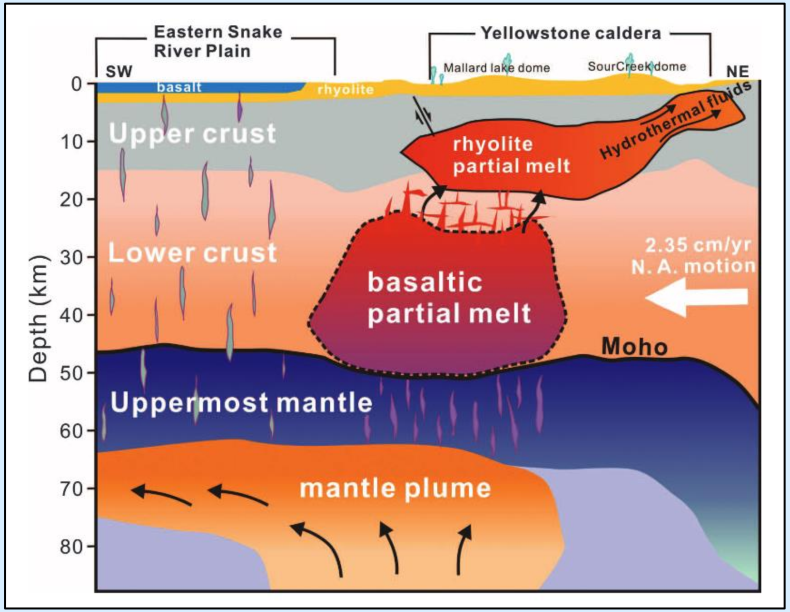
Two competing hypotheses

Fluid-driven unrest



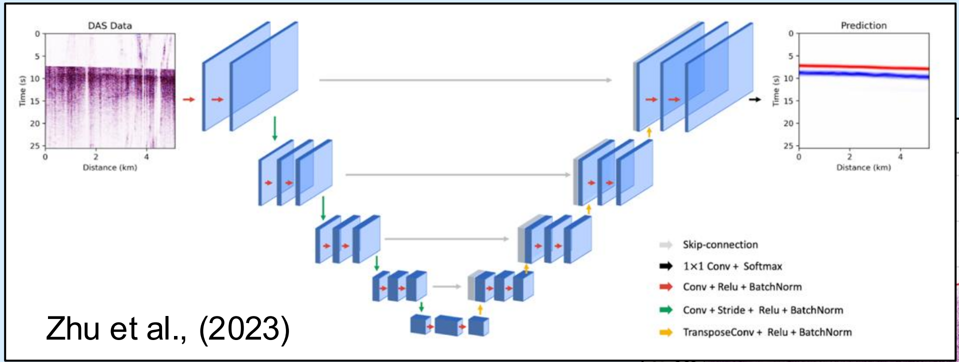
Hildreth et al., 2017

Upper-crust rhyolitic reservoir

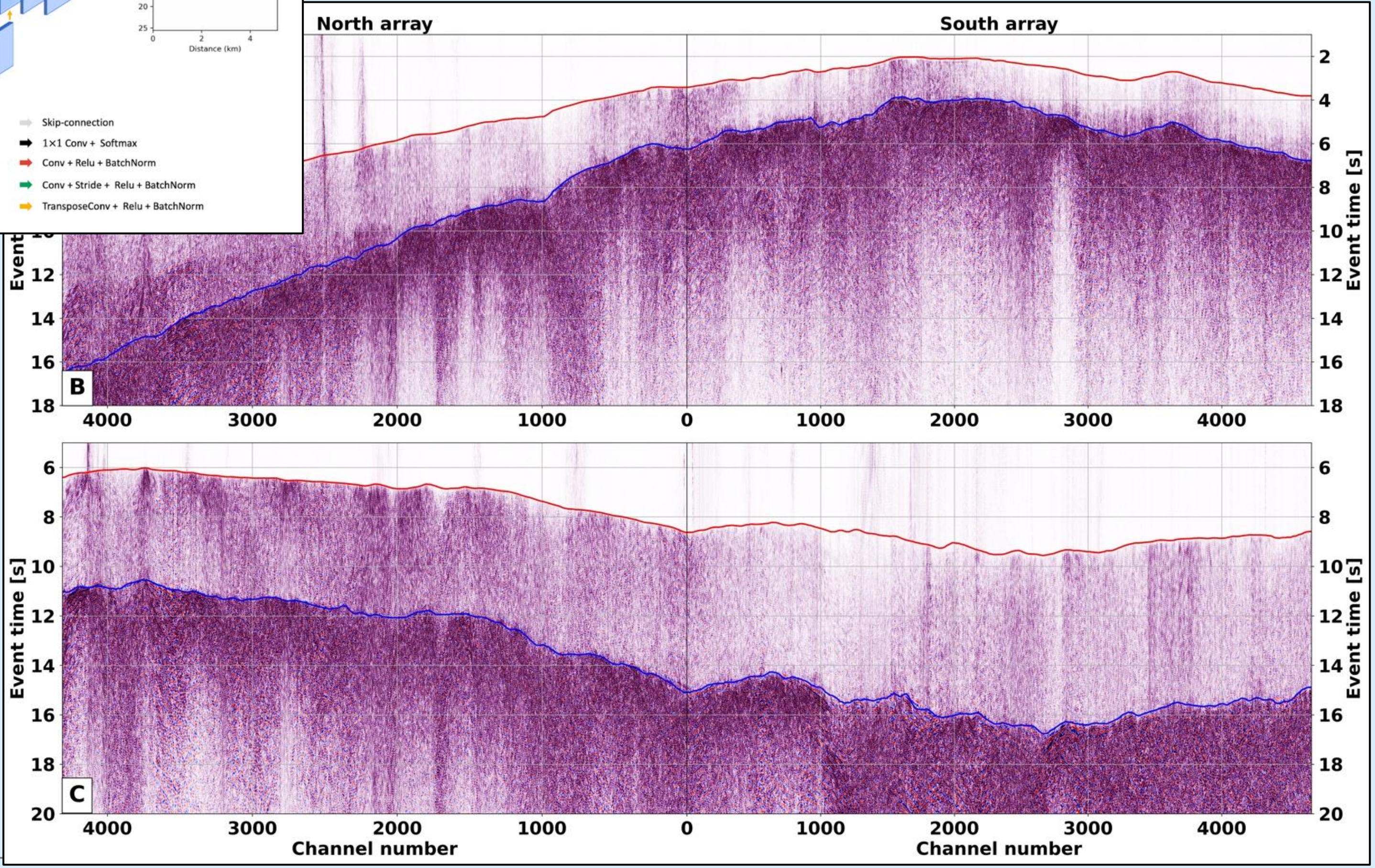


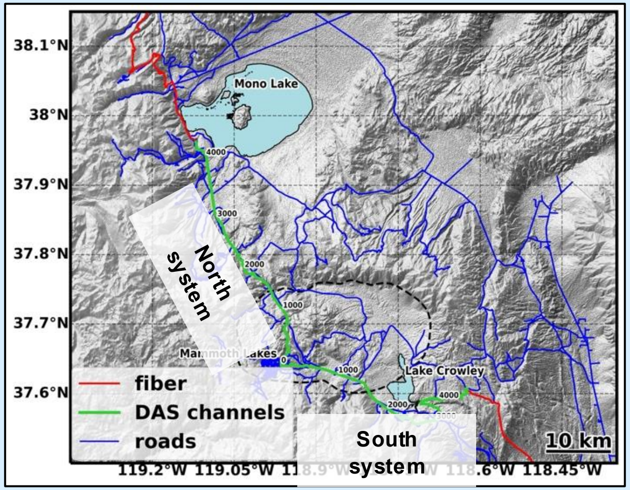
Huang et al., 2015

Accurate event picking: PhaseNet-DAS



Zhu et al., (2023)





3D tomography with DAS

Matrix-based approach

$$\boxed{\mathbf{Q}_{\mathbf{DD}}}\mathbf{A}\Delta\mathbf{X} + \mathbf{Q}_{\mathbf{DD}}\mathbf{C}\Delta\mathbf{M} = \mathbf{Q}_{\mathbf{DD}}\Delta\mathbf{T}.$$

$$\mathbf{Q}_{\mathbf{DD}} = \begin{bmatrix} 1 & -1 & \bullet & \bullet & \bullet & 0 \\ 1 & \bullet & \bullet & -1 & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ 0 & \bullet & 1 & \bullet & \bullet & -1 \end{bmatrix}$$

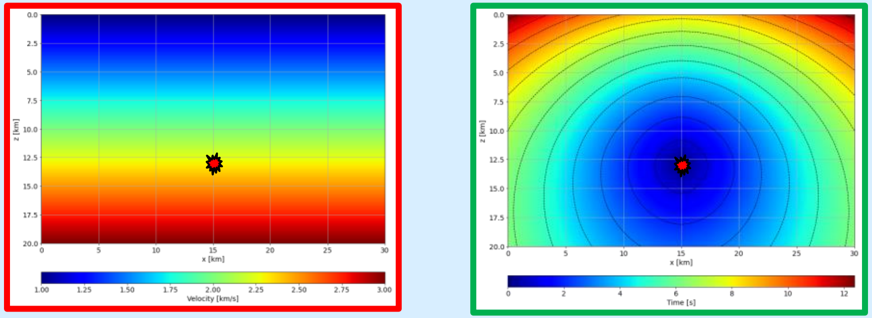
Common size $N \approx 10^5$

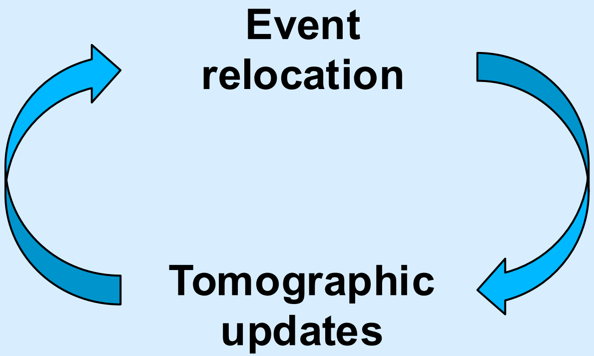
Size with DAS $N \approx 10^9$

Zhang and Thurber, 2006

Matrix-free strategy

$$|\nabla\boldsymbol{\tau}|^2 = \frac{1}{\boldsymbol{v}^2}$$





3D tomography with DAS

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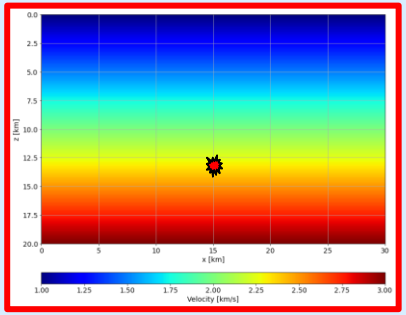
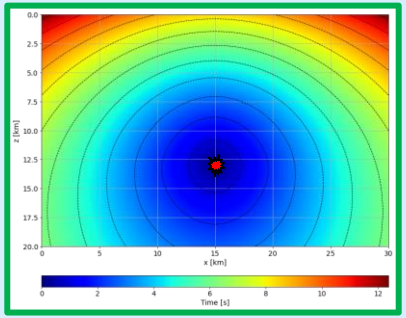
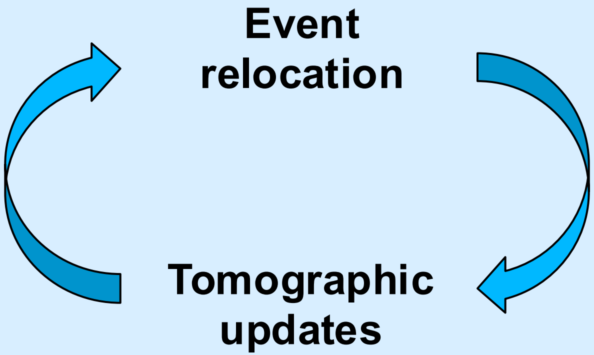






Operator-based workflow

$$|\nabla \boldsymbol{\tau}|^2 = \frac{1}{\boldsymbol{v}^2} \rightarrow \boldsymbol{f}(\boldsymbol{v}; \boldsymbol{x}_s, \boldsymbol{x}_r) = \boldsymbol{\tau}(\boldsymbol{x}_s, \boldsymbol{x}_r)$$

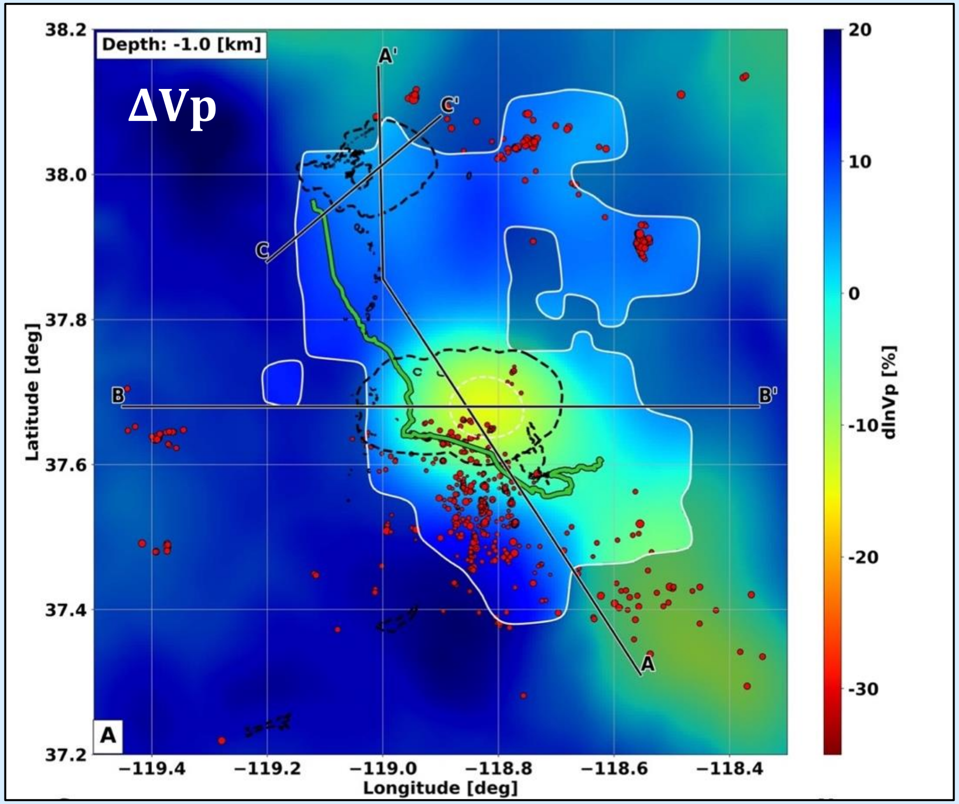
Eikonal non-linear operator

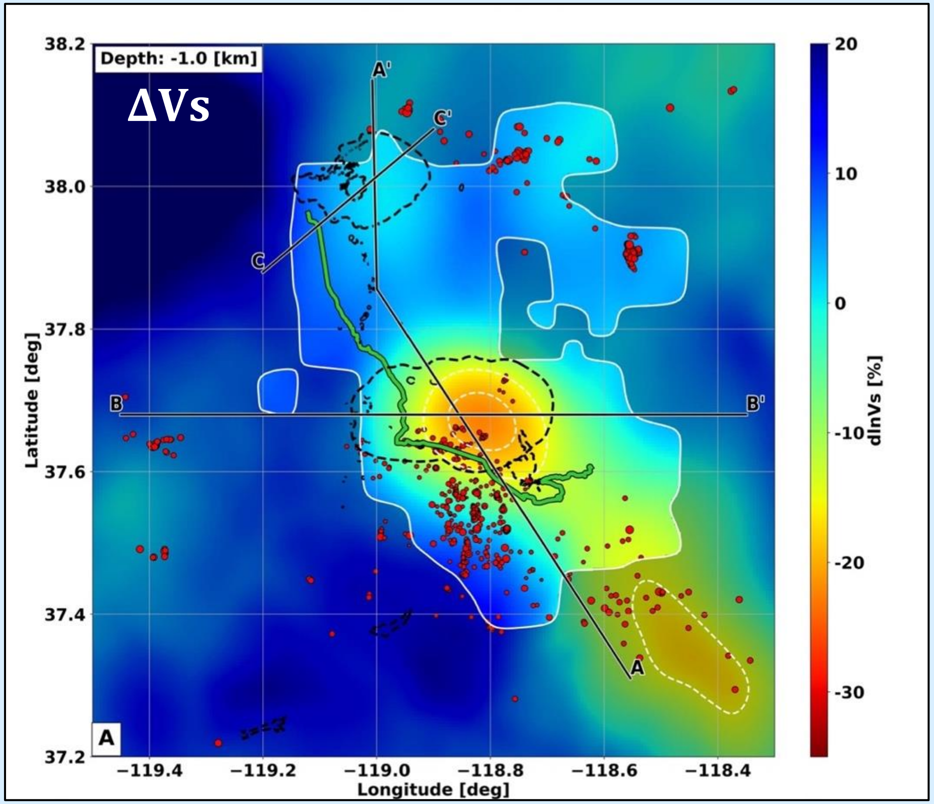
$$\phi(\boldsymbol{v}) = \frac{1}{2} \left| \boxed{\boldsymbol{f}(\boldsymbol{v}; \boldsymbol{x}_s, \boldsymbol{x}_r)} - \boldsymbol{\tau}_{obs}(\boldsymbol{x}_s, \boldsymbol{x}_r) \right|_2^2$$

Eikonal linearized operator

$$\nabla \phi(\boldsymbol{v}) = \boxed{\boldsymbol{F}^*(\boldsymbol{v})} [\boldsymbol{f}(\boldsymbol{v}; \boldsymbol{x}_s, \boldsymbol{x}_r) - \boldsymbol{\tau}_{obs}(\boldsymbol{x}_s, \boldsymbol{x}_r)]$$

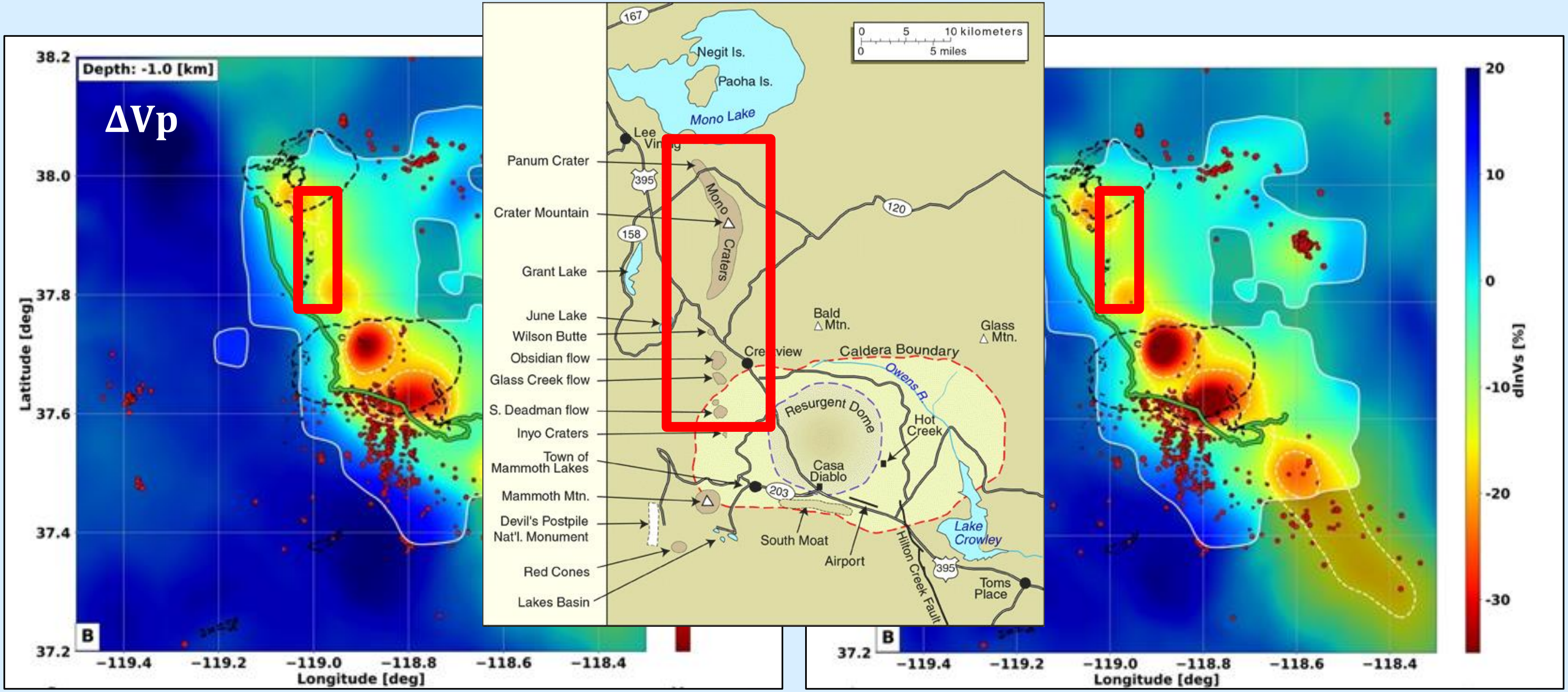
INITIAL MODELS

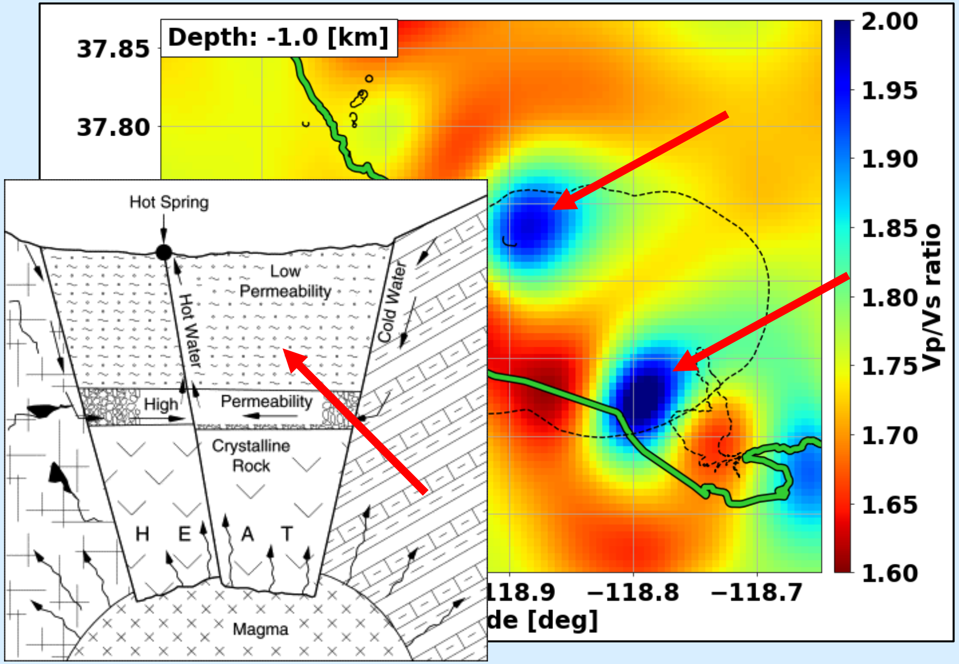




3D Eikonal Tomography: results

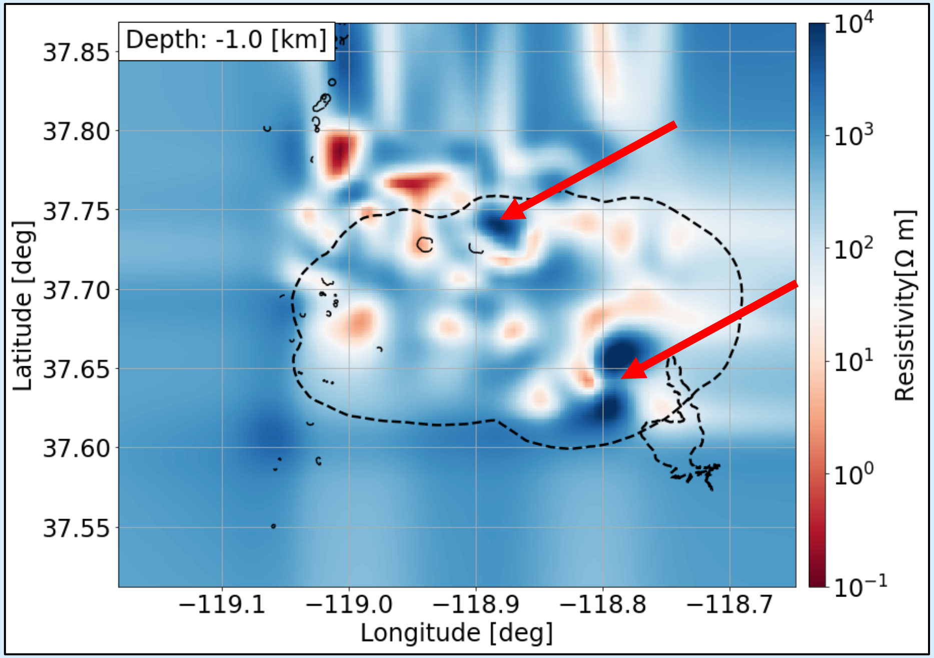
INVERTED MODELS





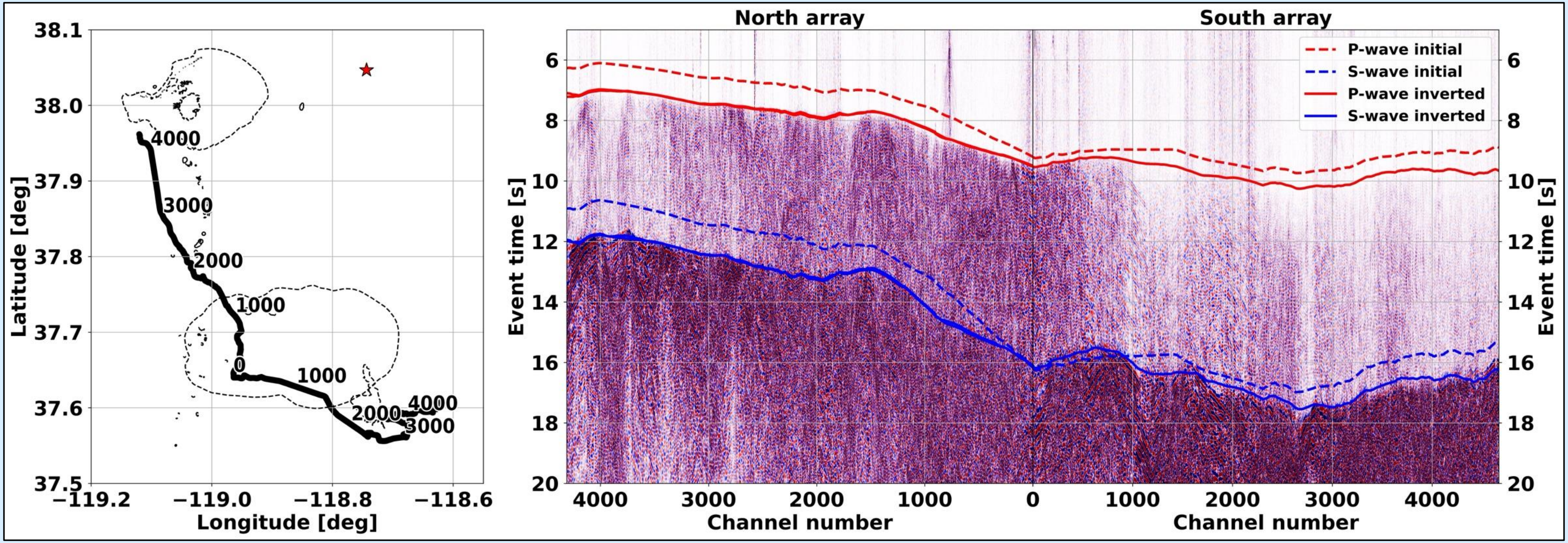
Cataldi et al., 1999

Biondi et al., 2023



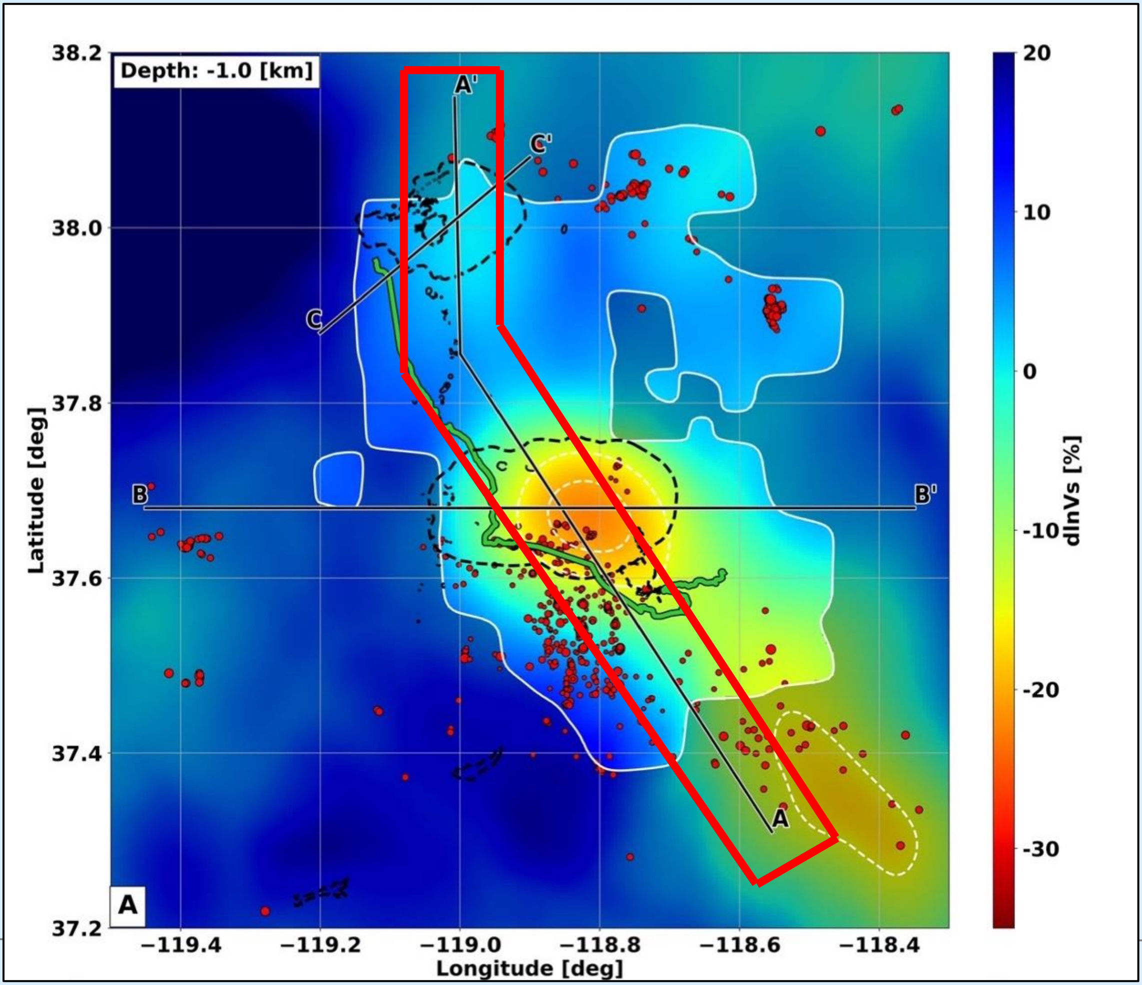
Peacock et al., 2016

3D Eikonal Tomography: results

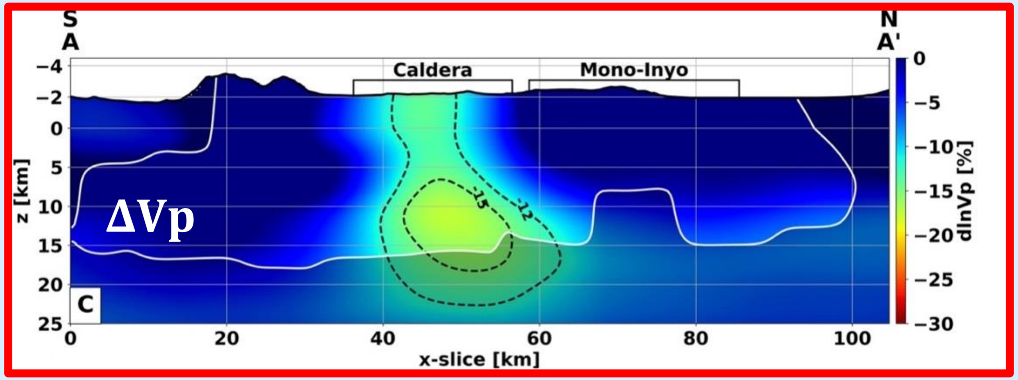


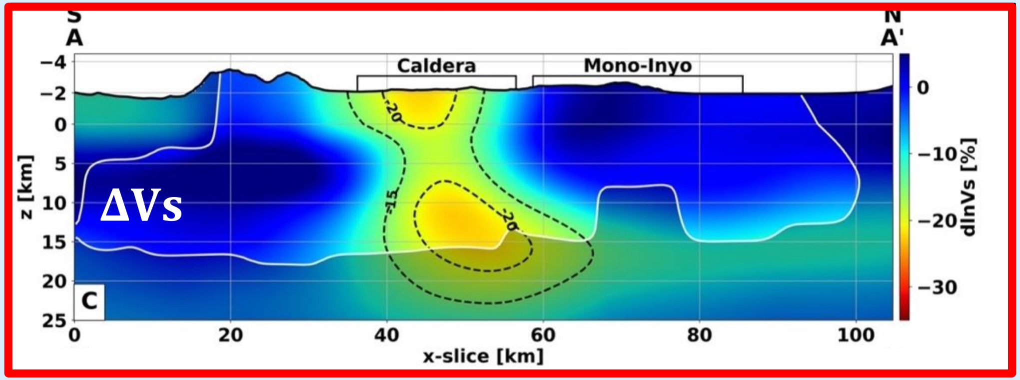
3D Eikonal Tomography: results

Let’s look at a few profiles!



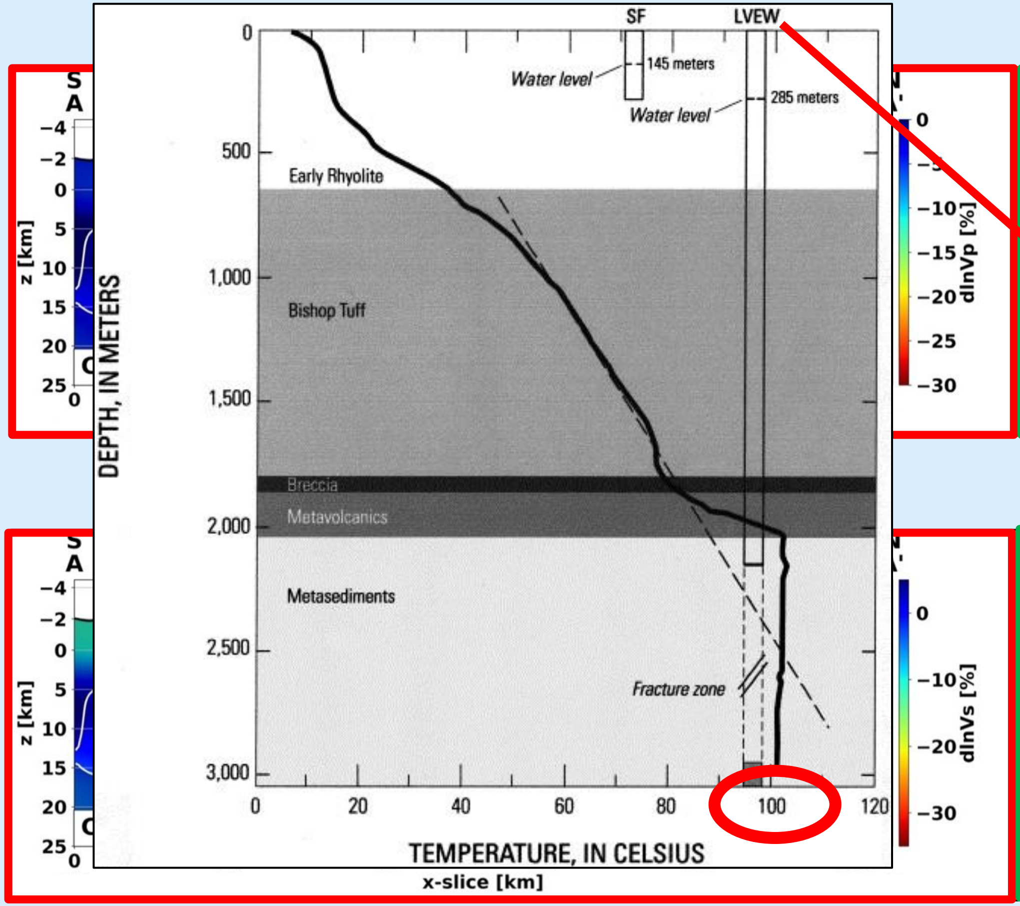
INITIAL MODELS

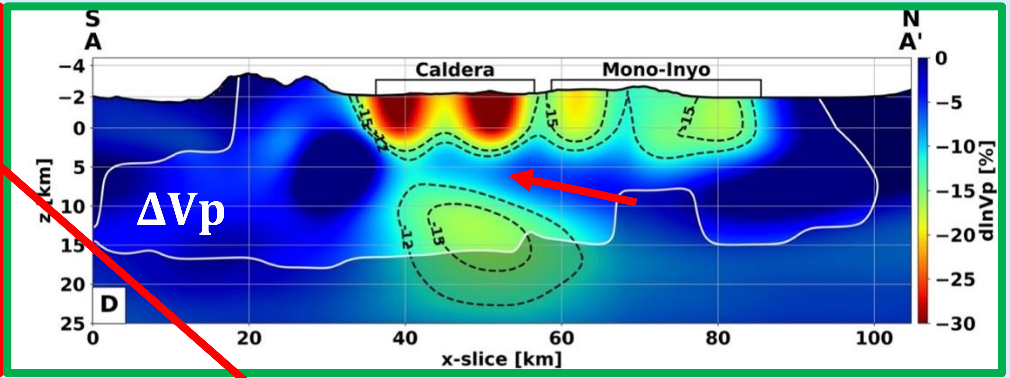


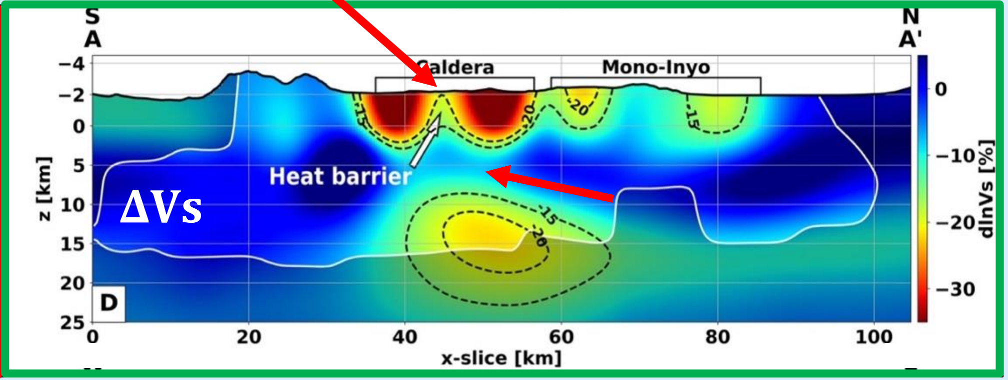


3D Eikonal Tomography: results

INVERTED MODELS

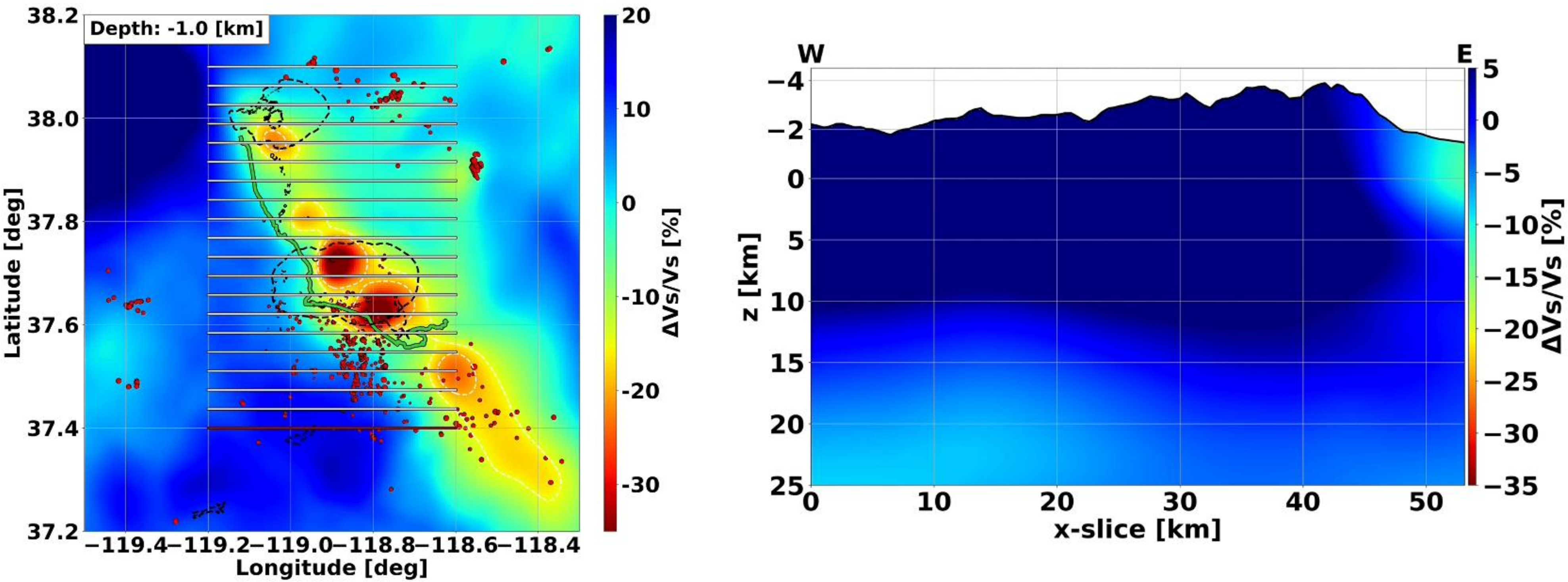




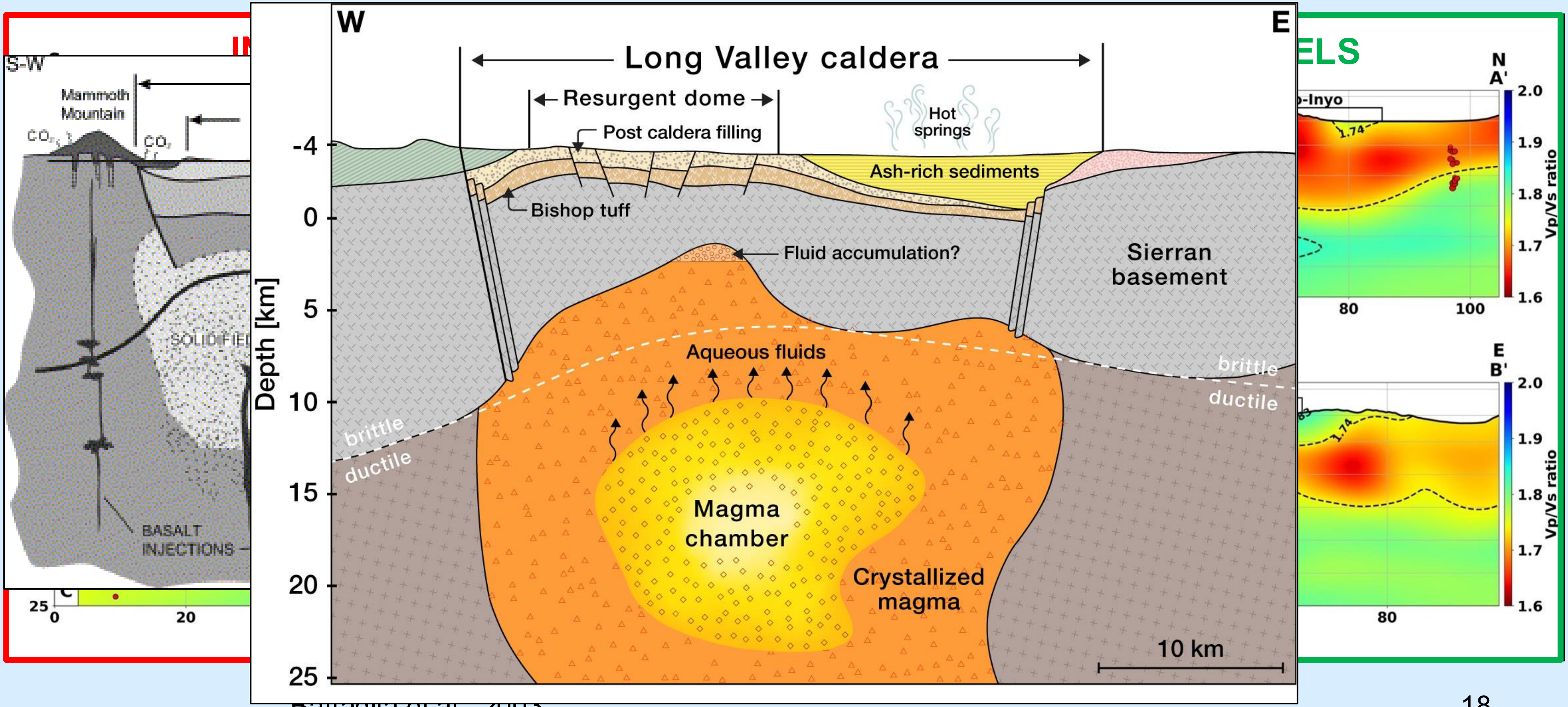


Sorey et al., 2003

3D Eikonal Tomography: results

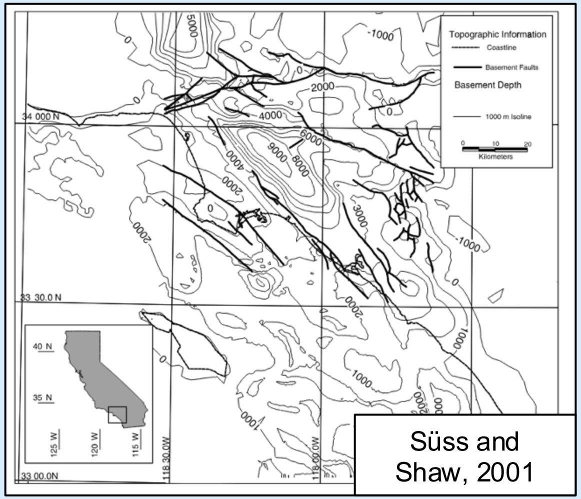


3D Eikonal Tomography: Vp/Vs

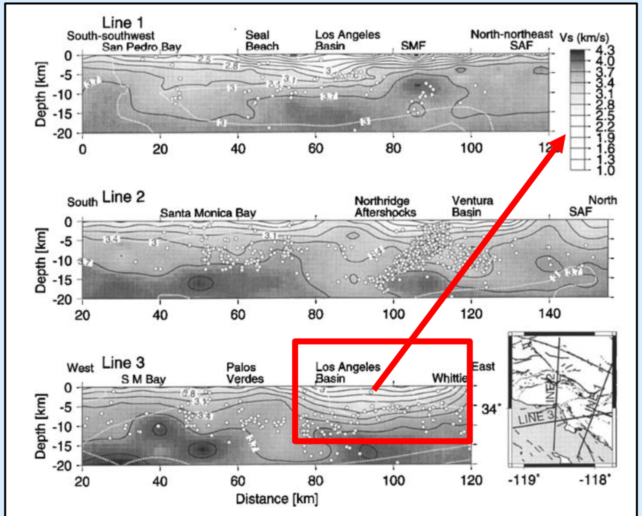


Battaglia et al., 2003

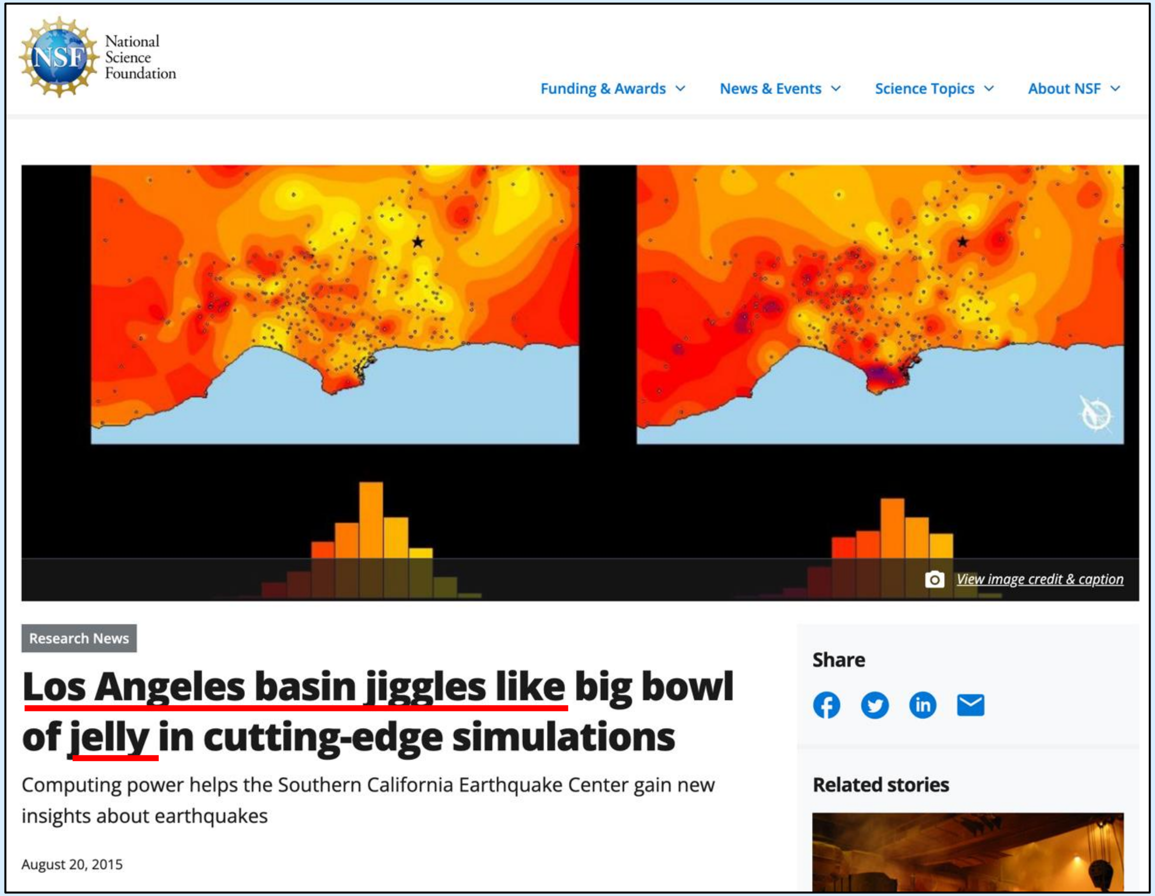
The Los Angeles basin



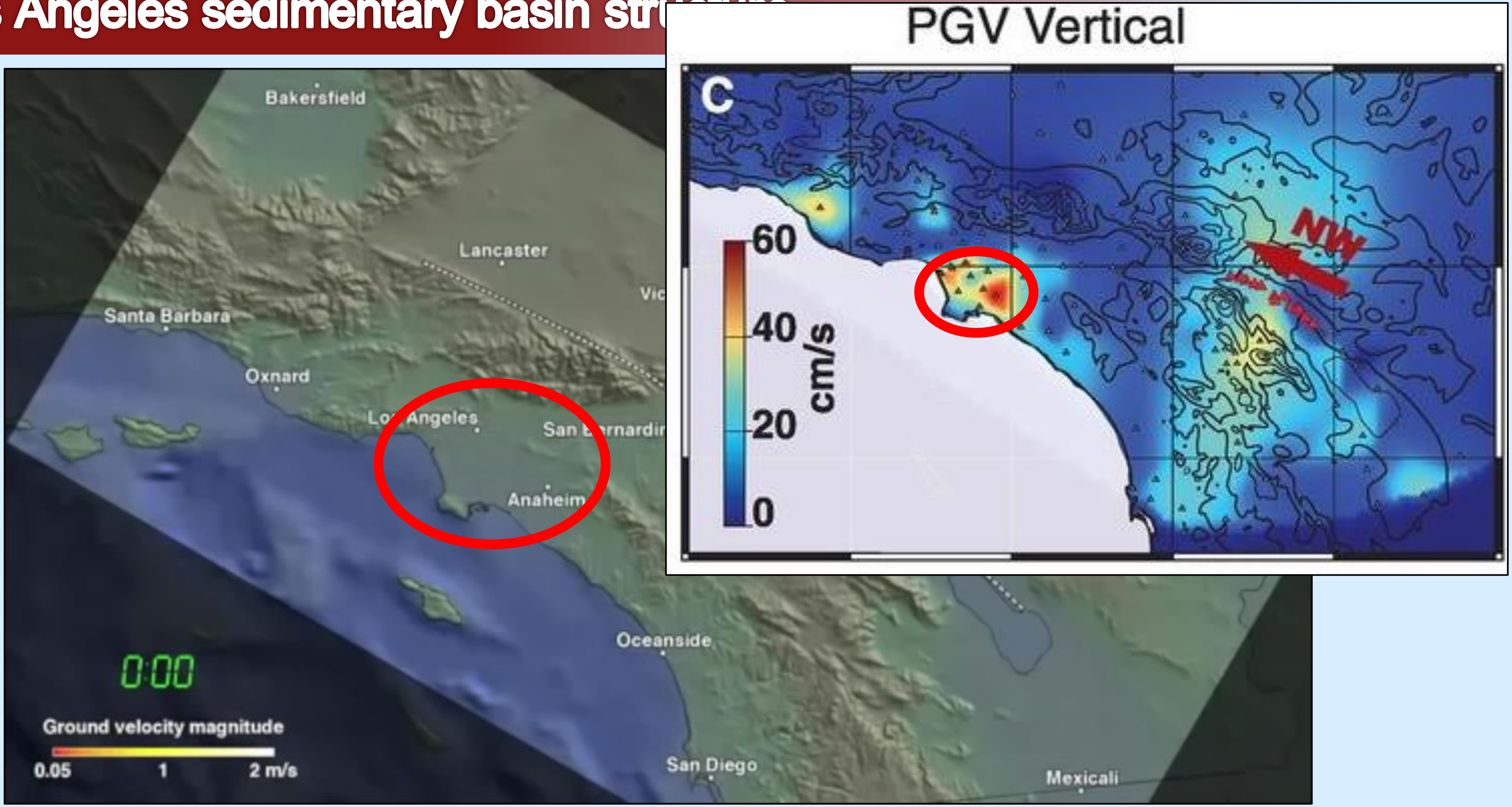
Süss and Shaw, 2001



Hauksson and Haase, 1997



Los Angeles sedimentary basin stru



Denolle et al., 2014

Community velocity models

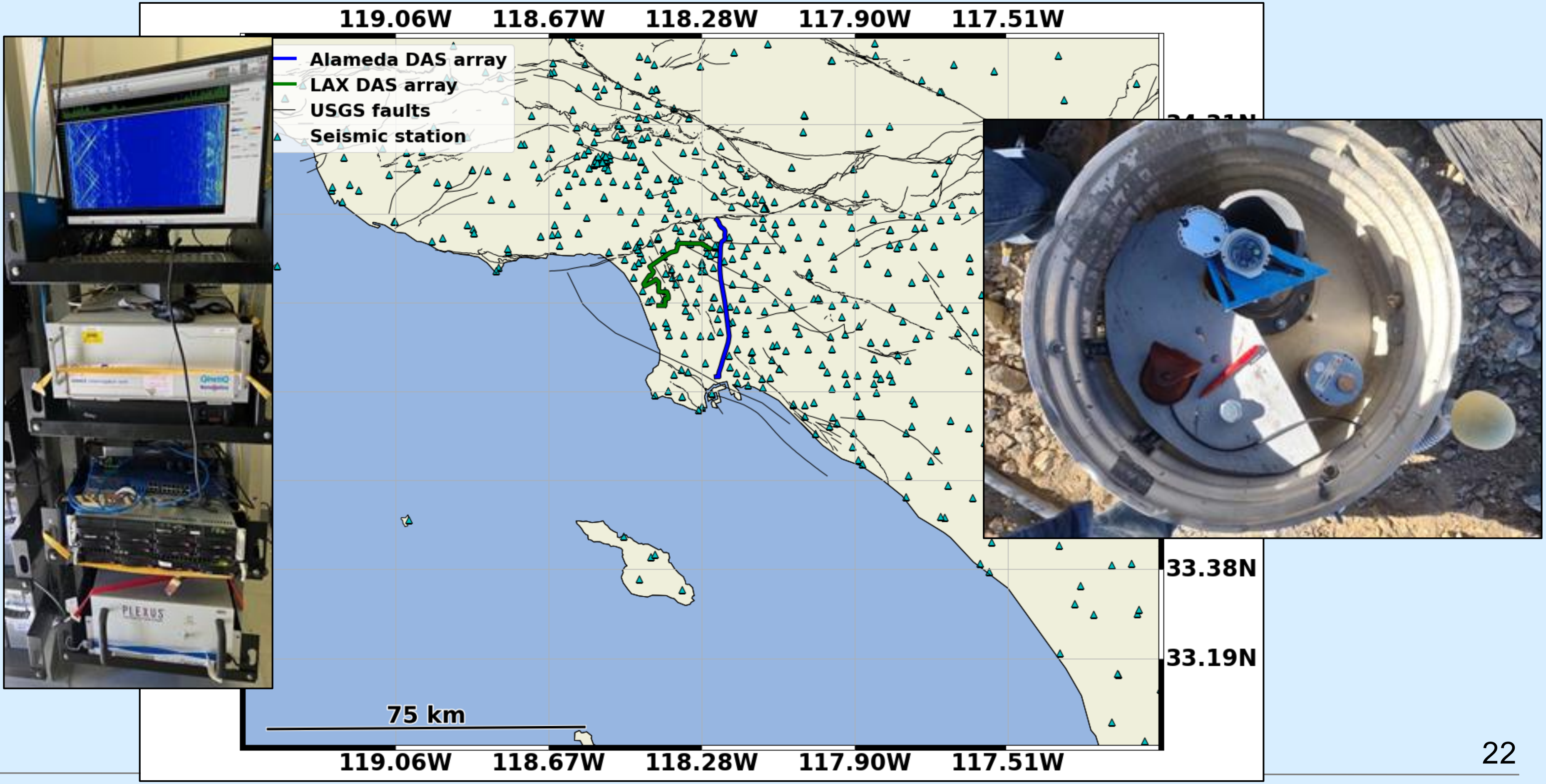
Small et al., 2017

Can we push the resolution of our current models further?

Characterizing the LA basin

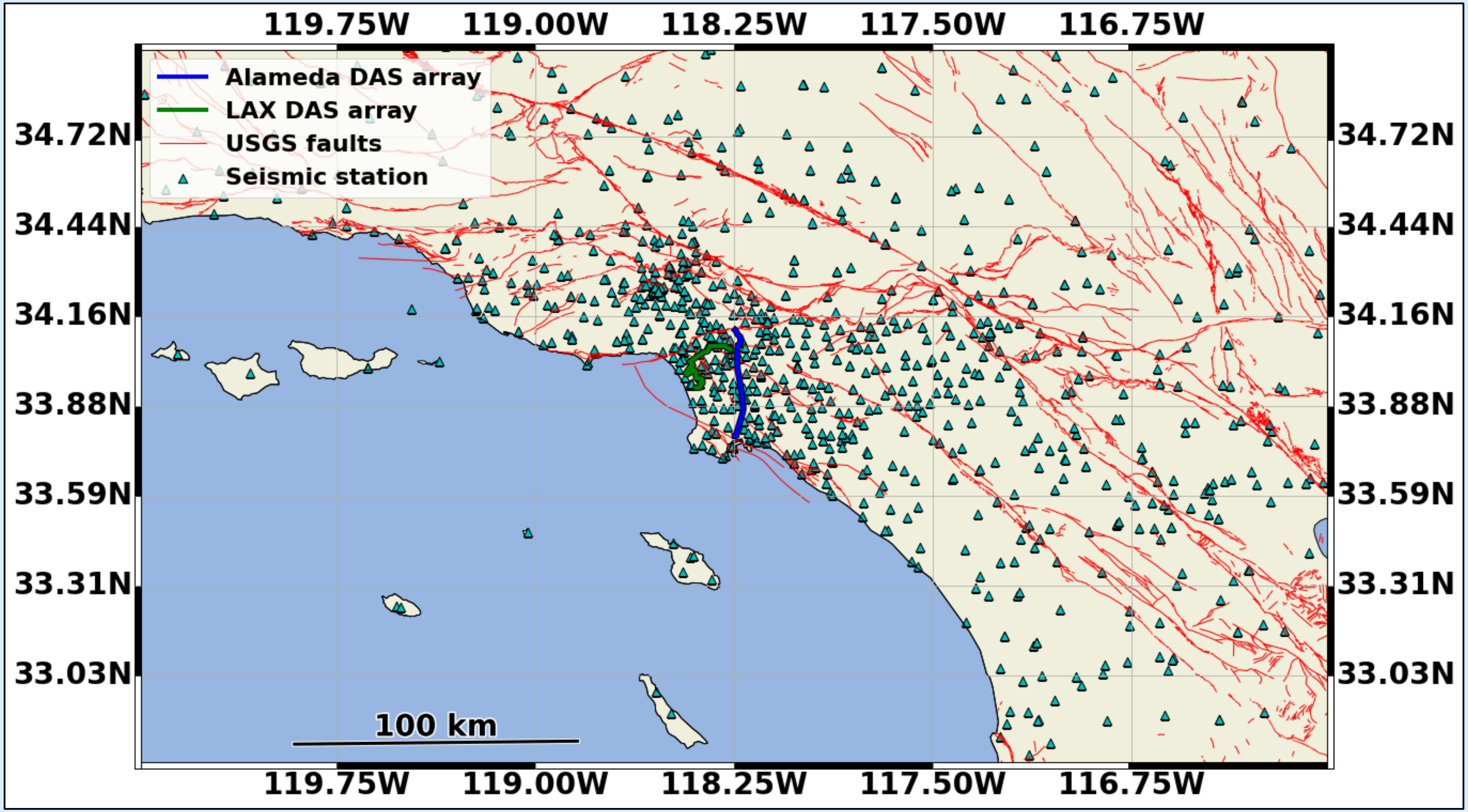
Distributed acoustic sensing

Seismic stations

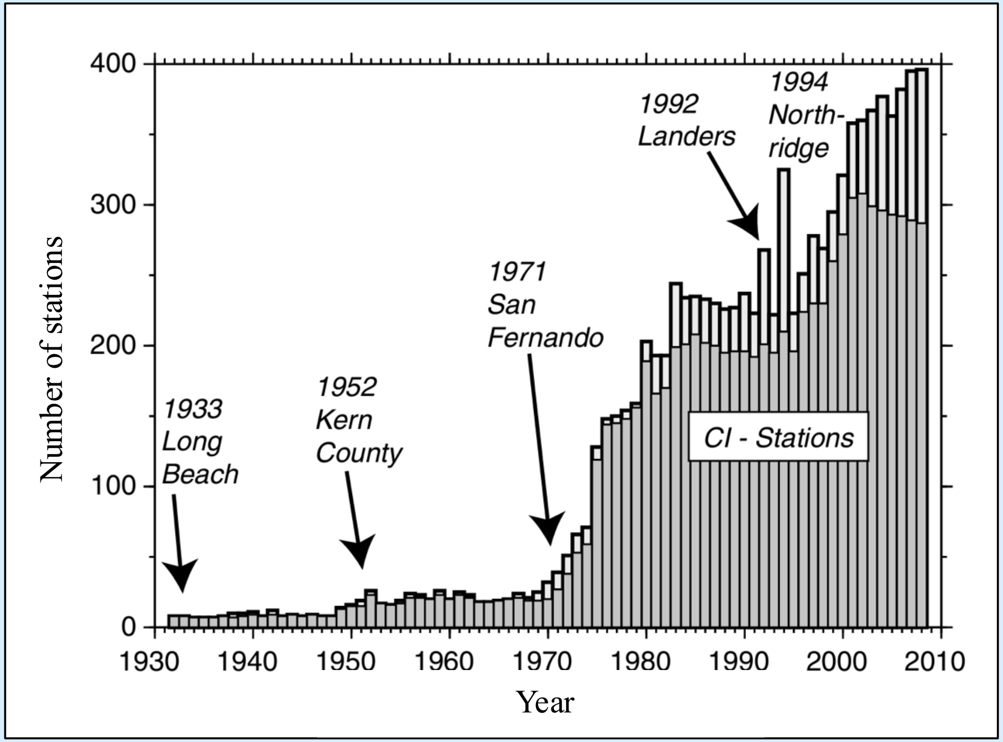


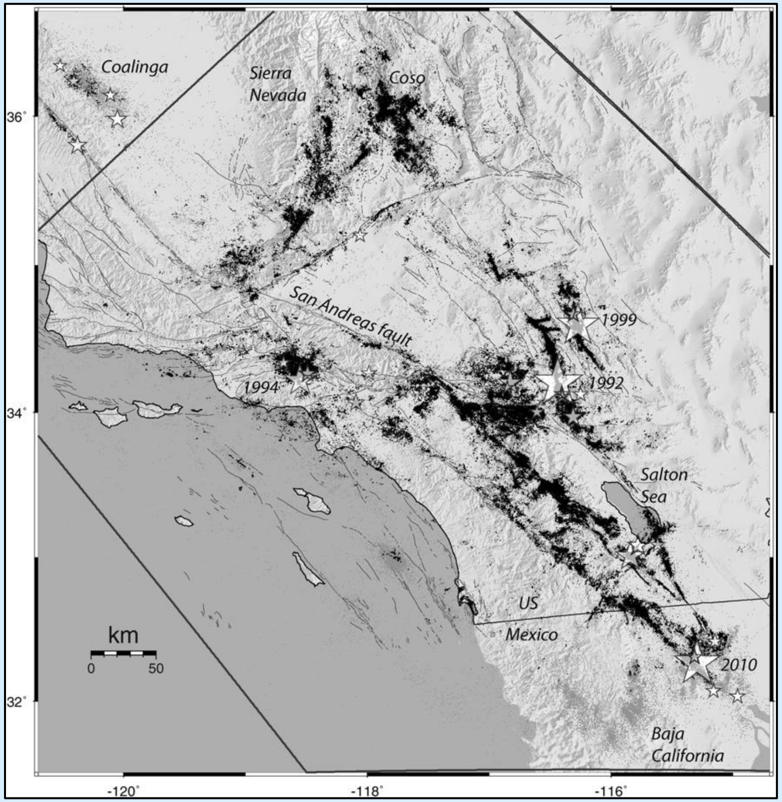
Leveraging existing and new data

More than 40 years of earthquake data!



Leveraging existing and new data



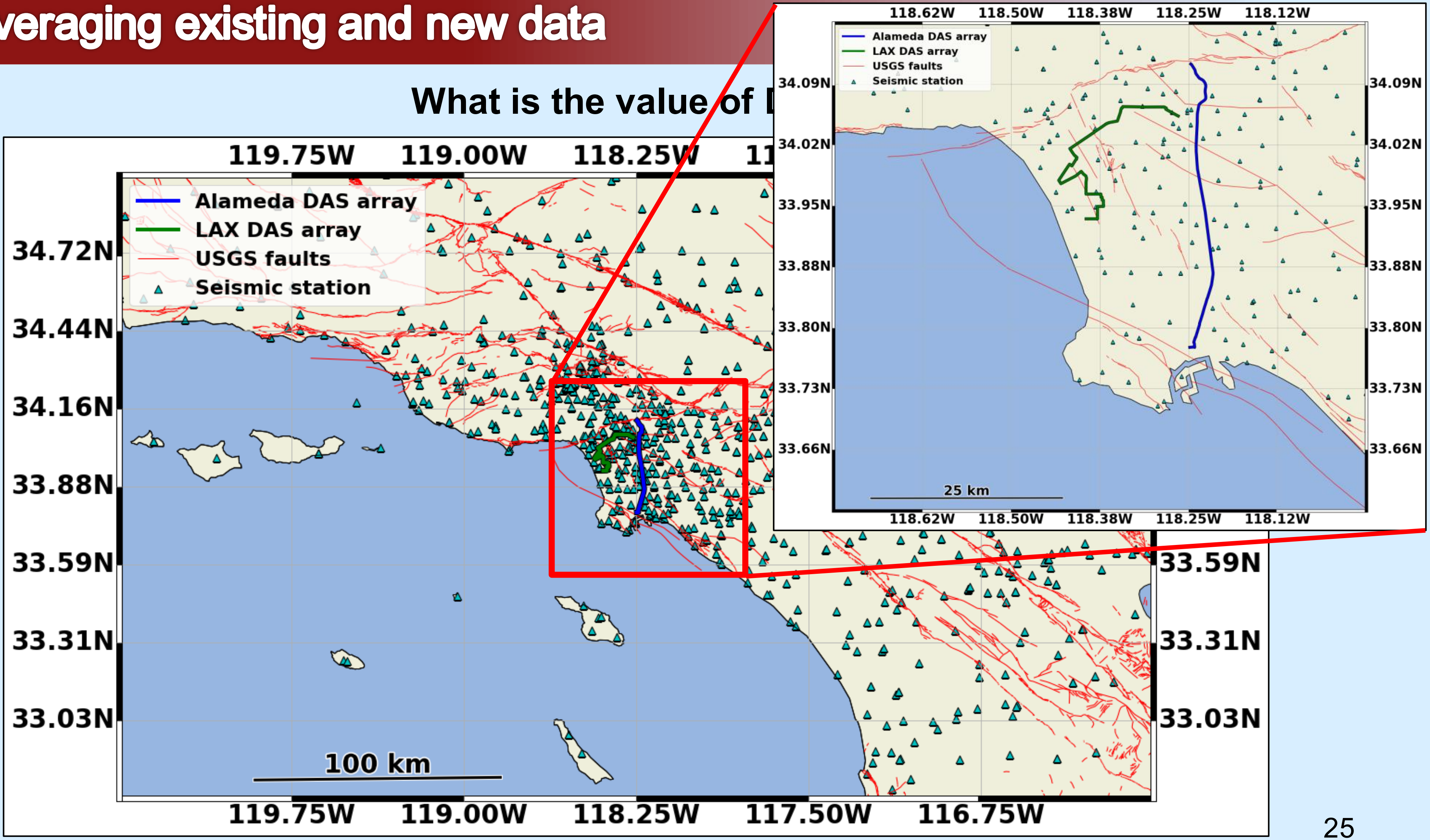


Hutton et al., 2010

Hauksson et al., 2012

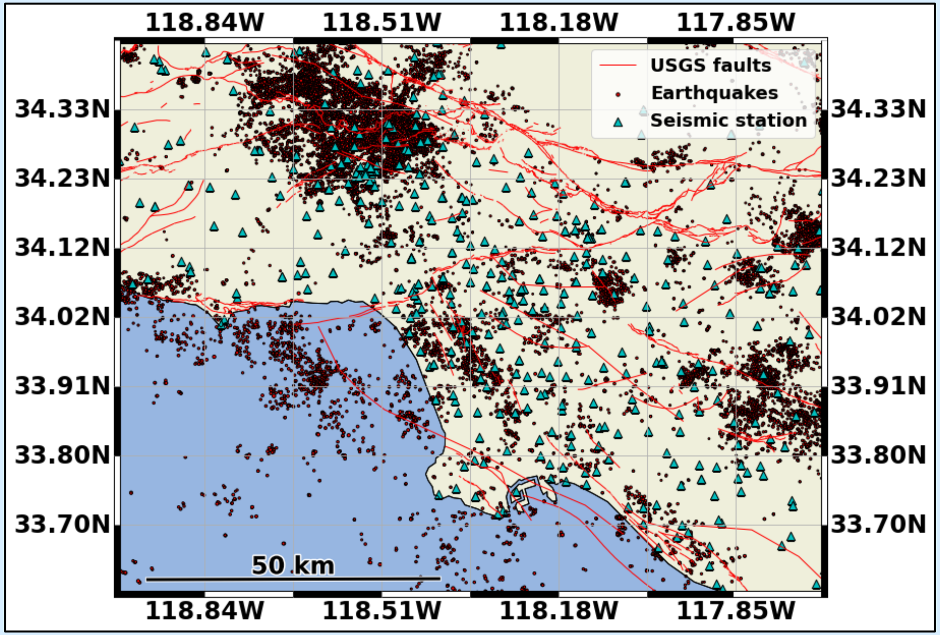
Leveraging existing and new data

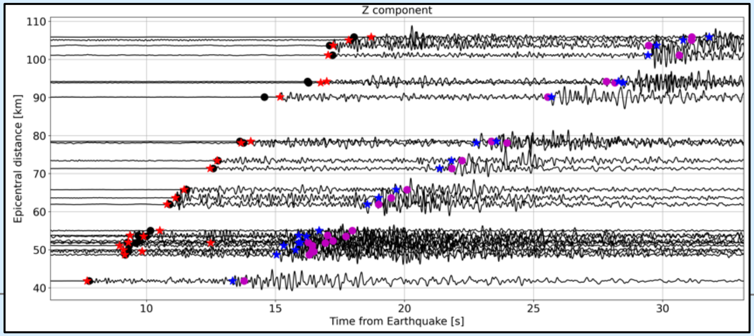
What is the value of D



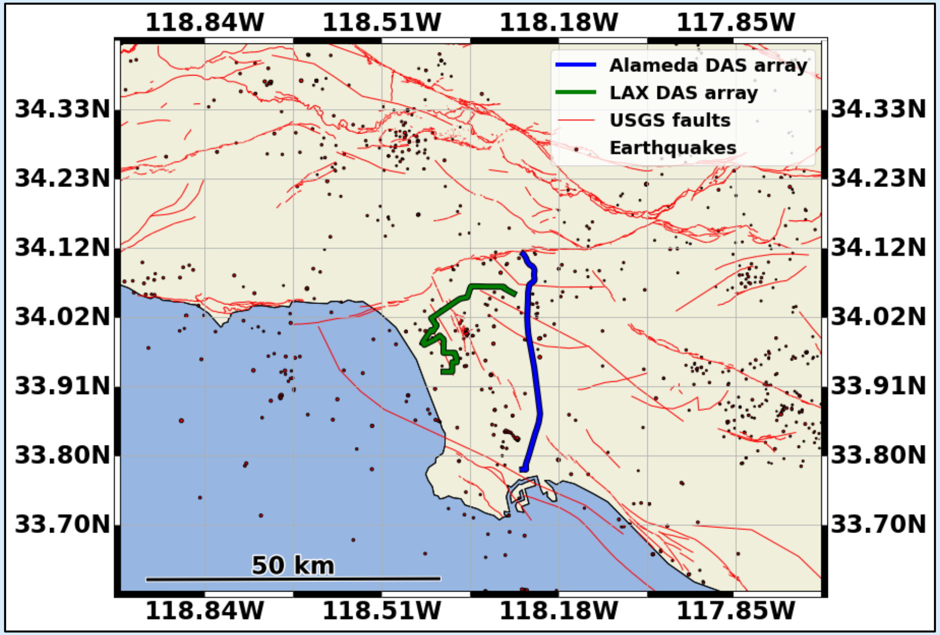
Leveraging existing and new data

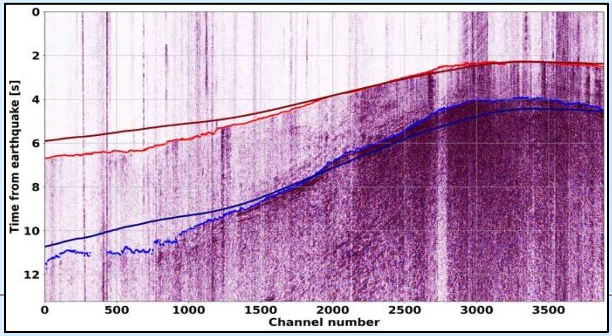
SCSN picks: 35,000 events since 1981





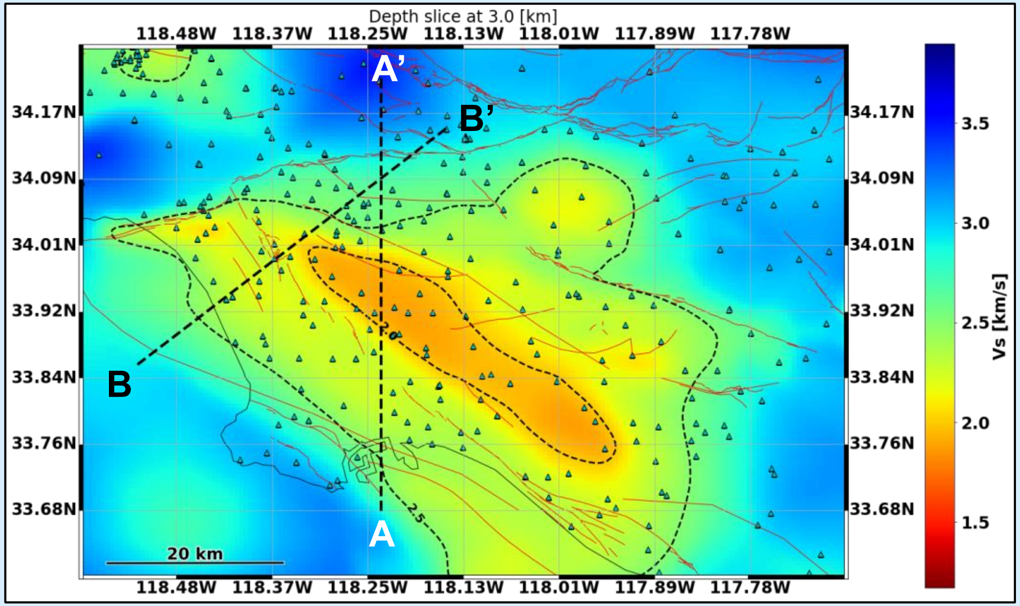
DAS: 1,000 events since 2022

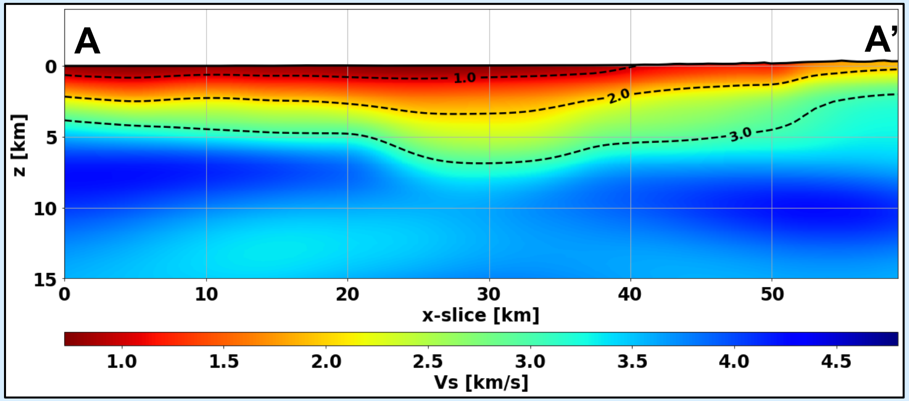


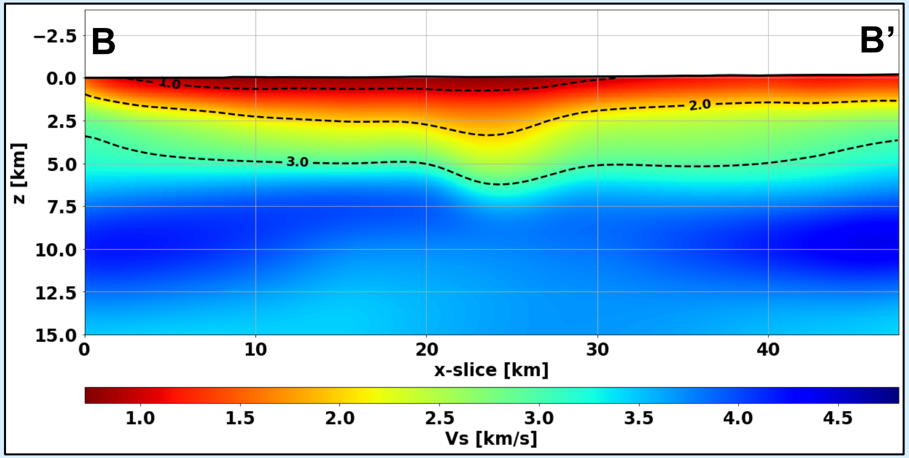


The Los Angeles basin: high-res tomography

Initial velocity model



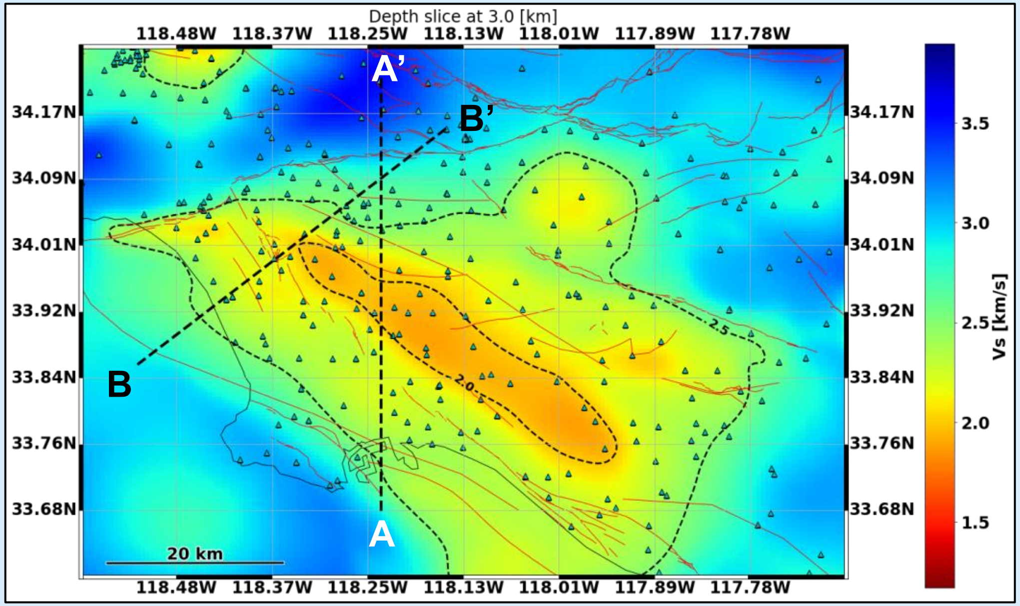


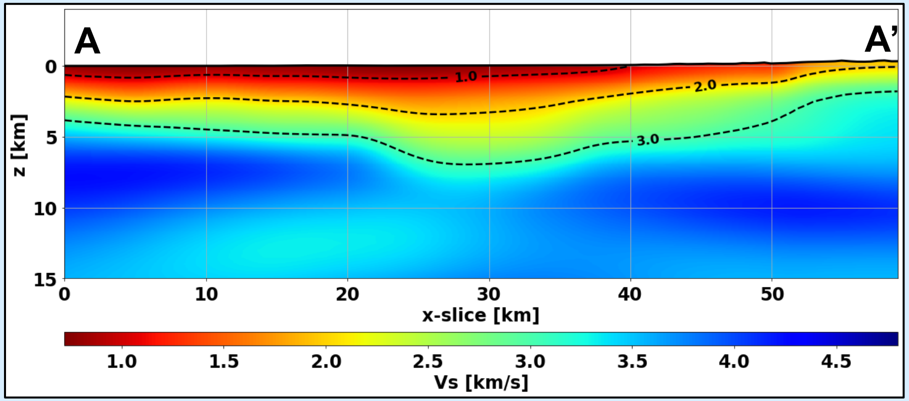


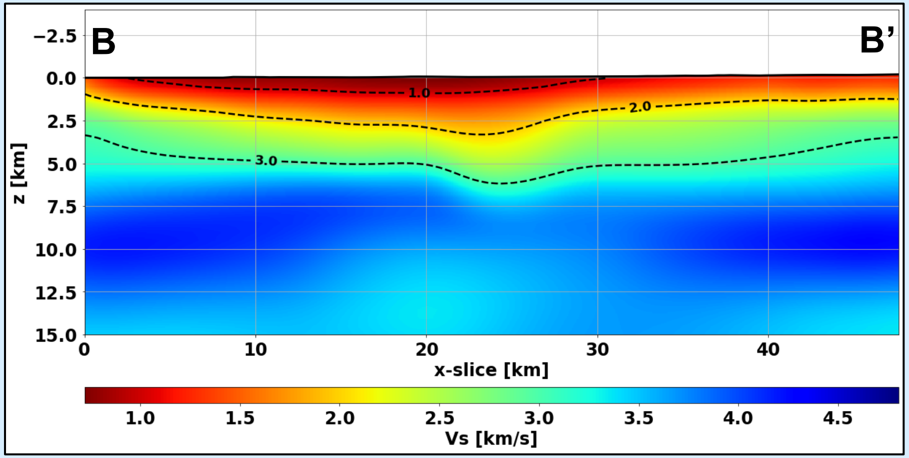
Small et al., 2017

The Los Angeles basin: high-res tomography

Inverted model from stations

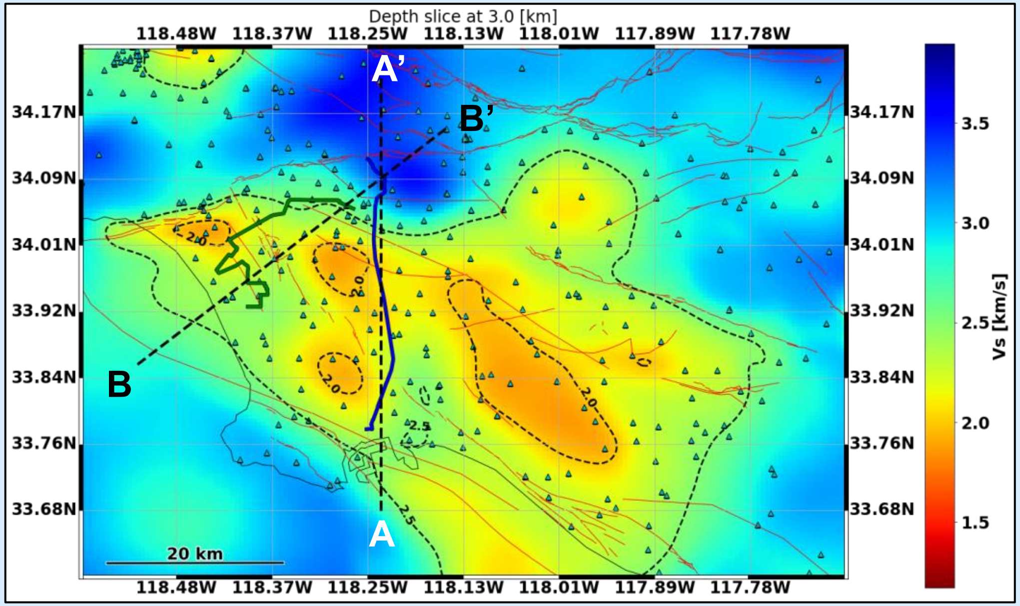


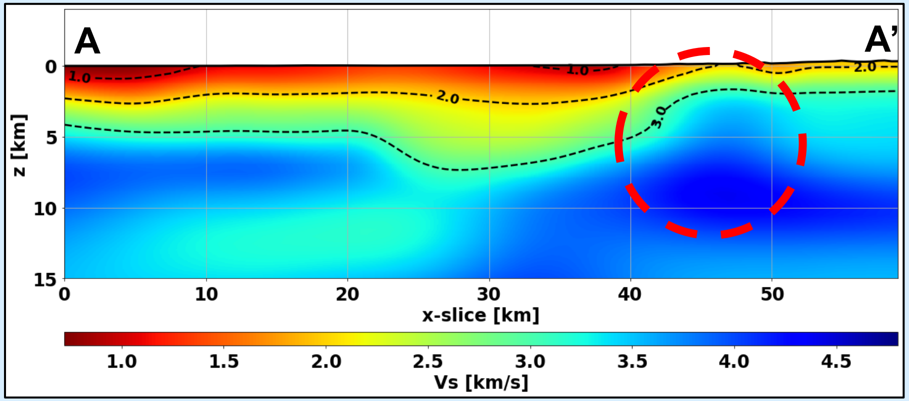


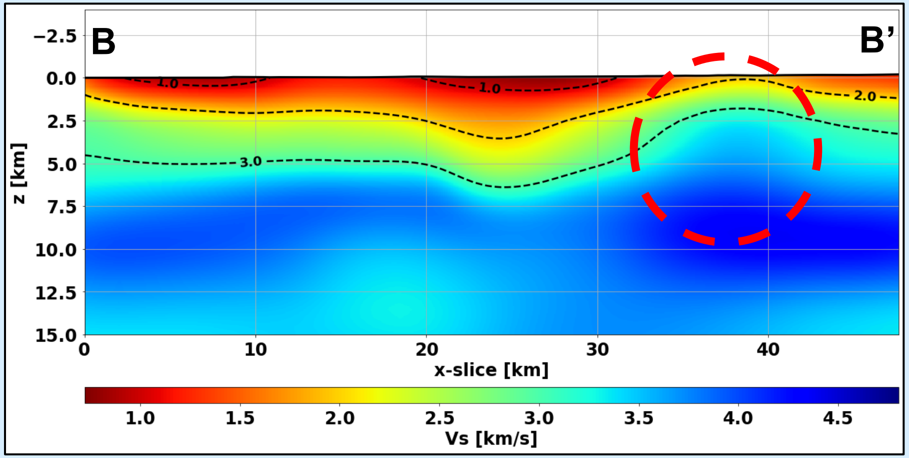


The Los Angeles basin: high-res tomography

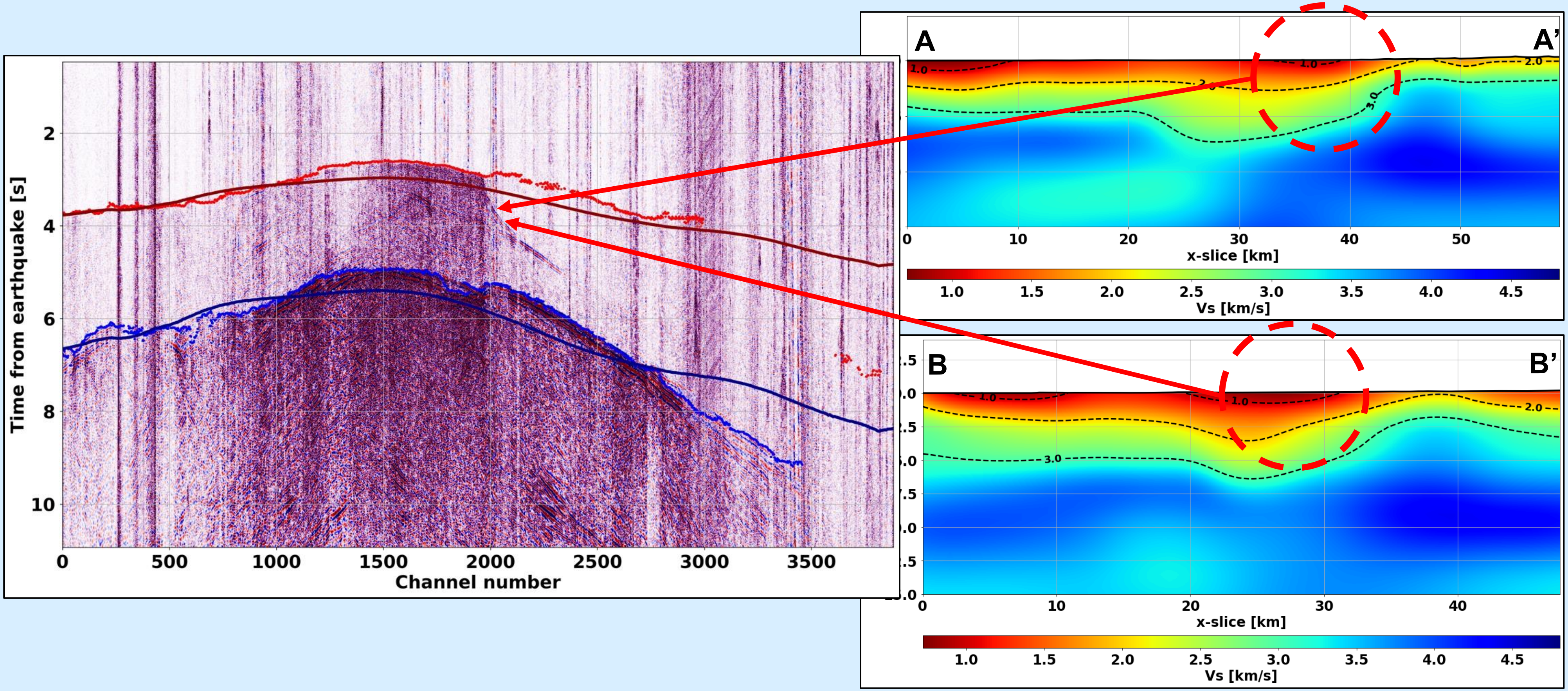
Inverted model from DAS



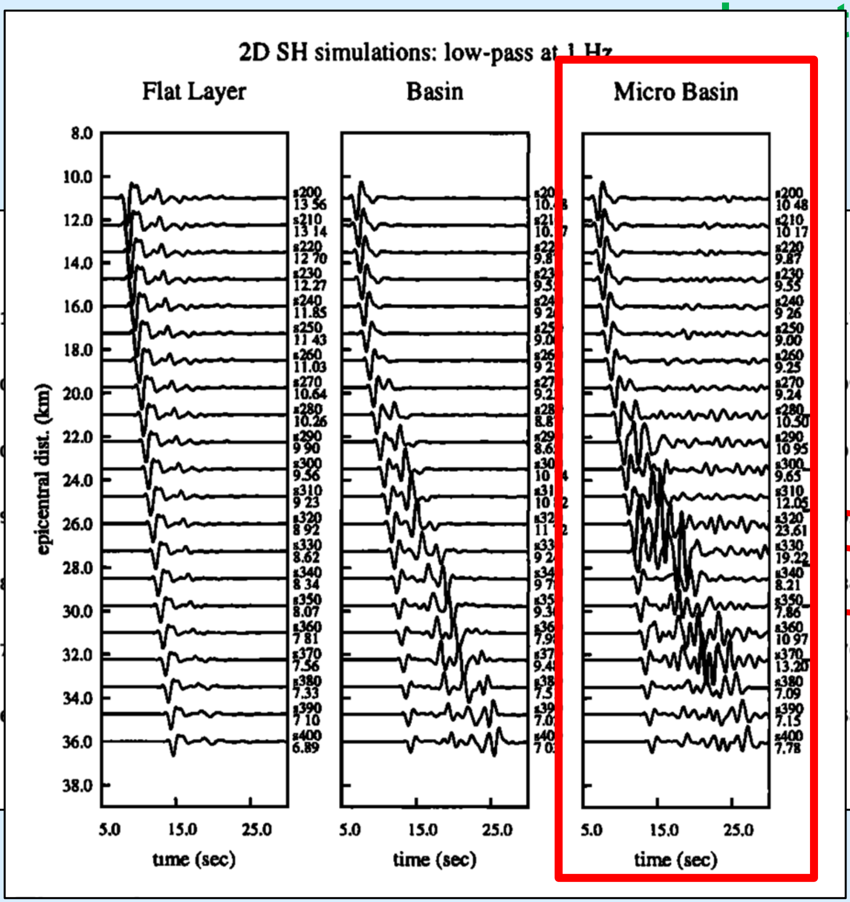


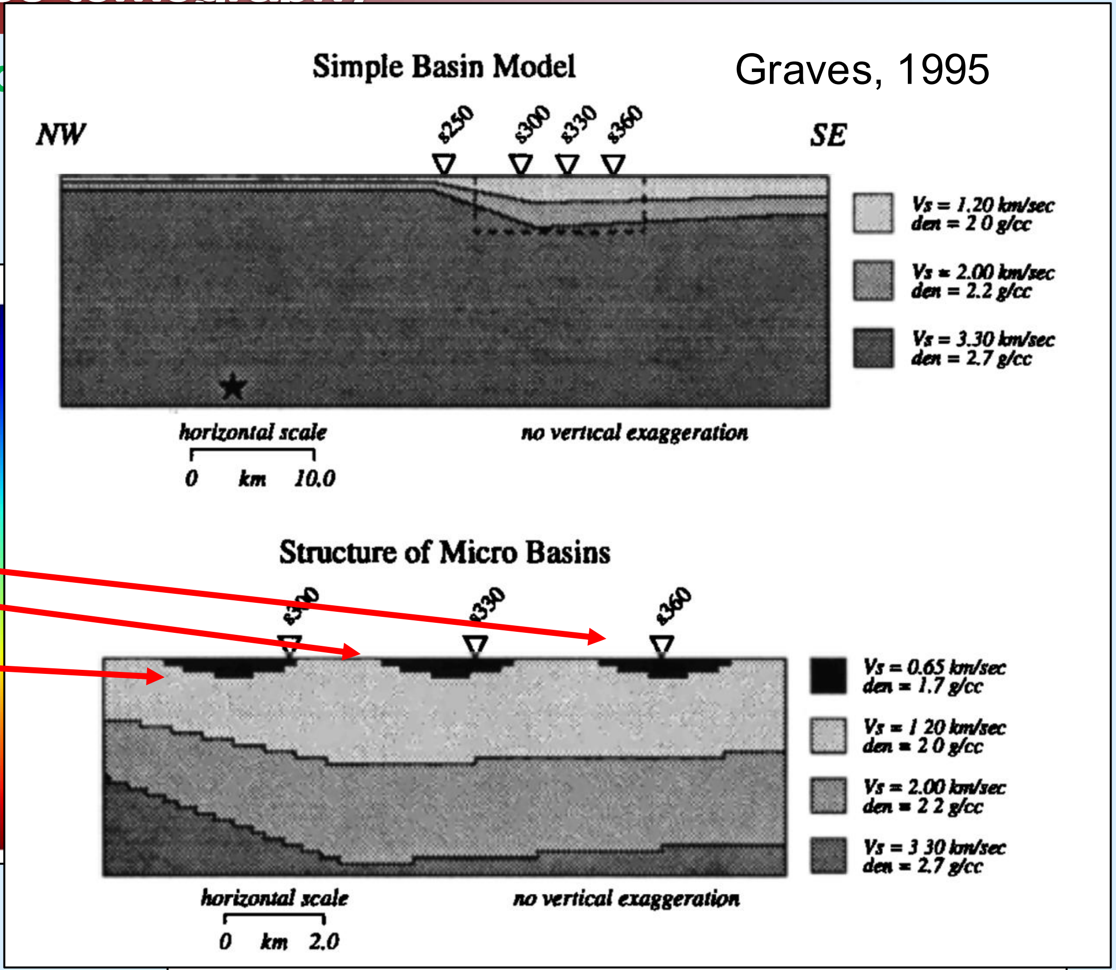


Inverted model from DAS



The Los Angeles basin: high-res tomography

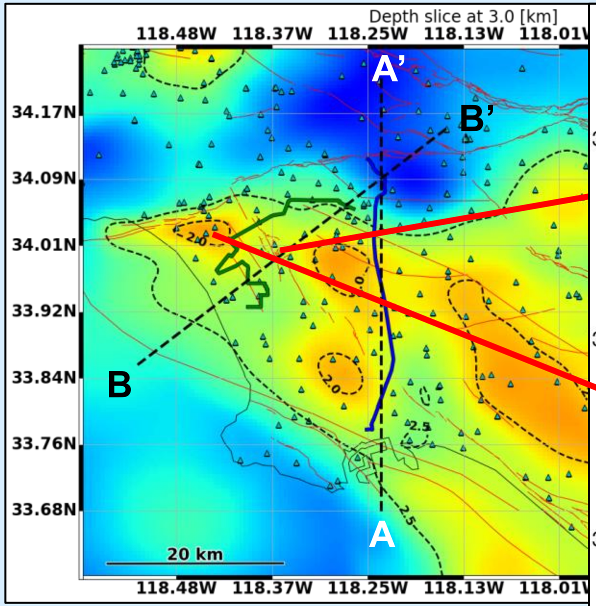


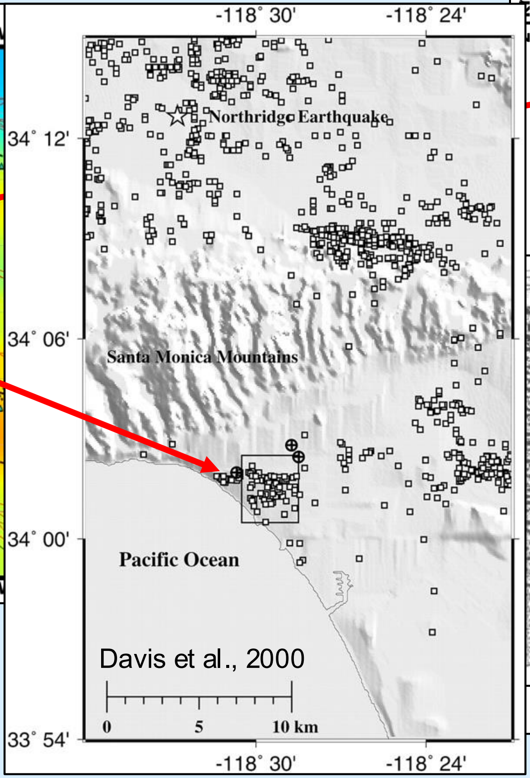


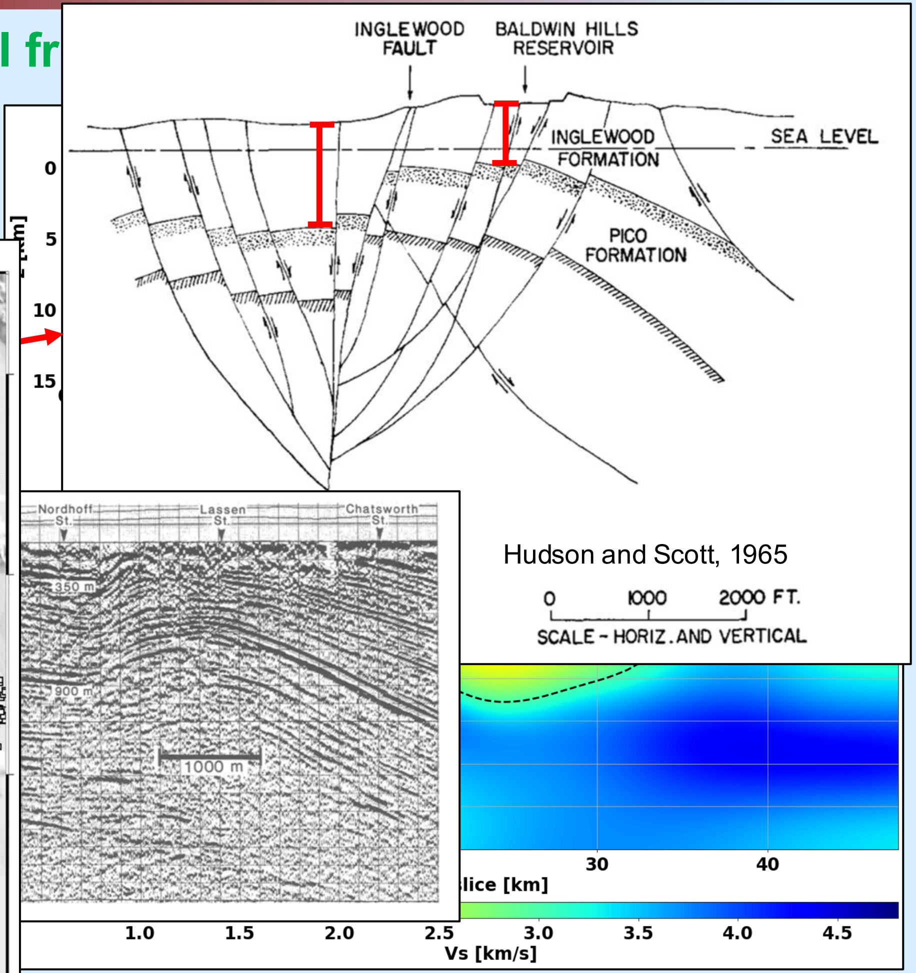
Vs [km/s]

The Los Angeles basin: high-res tomography

Inverted model fr







Conclusions

- **Distributed acoustic sensing (DAS) is providing a new tool to the seismological community to bridge multiple scales**

- **Seismic imaging is clearly benefiting from DAS by allowing us to reach a new resolution paradigm**

- **To harness a large-scale network of DAS instruments, we need novel processing workflows to provide useful data products**

Acknowledgements

- We would like to thank OptaSense for the constant support.
- We thank Google X for providing the LAX DAS data and Onward Network for the 395-network fiber access.
- We also thank the Los Angeles Department of Water and Power (LADWP) for providing access to their telecommunication fibers.

Thank you for your attention!
Questions?